Daylight Saving Time and Automobile Accidents: Evidence from Chile*

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Abstract

Under the evidence that the Daylight Saving Time (DST) regime does not accomplish its primary goal of saving energy, I analyze one of the main side effects, automobile accidents in Chile between 2002-2018. I use a Regression Discontinuity Design (RDD) exploiting the discrete nature of the transition into DST and a Difference-in-Difference (DID) approach, taking advantage of the changes in dates that the policy starts and ends over the years. I find a 2.7% reduction in automobile accidents under the DST regime. I isolate the two main mechanisms: sleep disruption and the reallocation of light. I find suggestive evidence that the sleep disruption effect plays a relevant role at both transitions: it increases automobile accidents by 6% the first week following the transition into DST and decreases them by 3.9% the first week following the transition into Standard Time (ST). I also find evidence that ambient light reduces serious and fatal accident risk.

Keywords: Daylight Saving Time, Automobile Accidents, Circadian Rhythm, Light Conditions

JEL Codes: R41, I12, I18, Q48

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1 Introduction

Daylight Saving Time (DST) was conceived as an energy-saving-based policy. Under this policy, every spring (fall), the clocks are set forward (backward) by one hour, commonly referred to as Daylight Saving Time (Standard Time, ST). In recent years, the literature has shown that DST policy does not save energy (Kellogg and Wolff 2008, Kotchen and Grant 2011, Choi et al. 2017, Verdejo et al. 2016, Kudela et al. 2020)[1]. Despite these findings, DST transition is still implemented in 70 countries across the world[2]. In this paper, I focus on one important indirect effect: the impact of the DST policy on automobile accidents.

To investigate this question, using detailed administrative data from Chile from 2002 to 2018 and the methodology used by Smith 2016[3], I measure the impact of a shock that affects the population twice yearly on the number of automobile accidents in the short, medium, and longer-run. It has been hypothesized that two main mechanisms can impact automobile accidents. First, the disruption of the circadian rhythm affects the sleep patterns the days following each transition[4]. Second, the reallocation of light from the morning (evening) to the evening (morning) creates riskier (safer) mornings and safer (riskier) evenings in the transition into DST (ST), and at the same time, all other activities, including commuting stay the same[5].

This question is relevant for a variety of reasons. First, the literature has found limited or non-existent energy savings as a result of DST implementation. For Chile, Verdejo et al. 2016 showed that DST has a small positive impact on energy savings. Therefore, if the DST transition impacts road accidents, it is important to assess the impact to guide the public policy discussion. Secondly, the literature reports mixed and non-conclusive evidence on the impact of the DST policy on automobile accidents. Previous research focusing on short-run consequences has found either an increase (Smith 2016, Goodwin et al. 2023), decrease (Varughese and Allen 2001, J. M. Sullivan and Flannagan 2002, Laliotis et al. 2023) or no effect (Lahti et al. 2010, James 2022) on automobile accidents. Other papers focusing on the long-run consequences found either a reduction (Sood and Ghosh 2007, Bünnings and Schiele 2020) or no effect (Smith 2016, James 2022) on automobile crashes under the DST regime. Thirdly, automobile accidents affect many people, regardless of their age or labor status.

Automobile accidents also do not only have direct costs (damage to property, medical costs,

[1] Balia et al. 2023 point out that recent papers have shown that DST increases the use of heating and air-conditioning following the transitions offsetting the decrease in energy use for lighting. Other papers have found negligible energy savings. They argue that nowadays, in the policy debate, it is relevant to consider this due to concerns about climate change, rising temperatures, and worldwide increases in cooling usage during summer

[2] DST has been scrutinized recently. For instance, in March 2019, the European Parliament voted for the abolition of seasonal clock changes by 2021, with member states keeping their right to decide on their time zone after that year.

[3] This methodology was firstly developed by Sexton and Beatty 2014 and Doleac and Sanders 2015, and then implemented for fatal automobile accidents by Smith 2016

[4] It is recommended to sleep between 7-8 hours per night. According to the Time Use Survey for Chile in 2015, the population between 25-65 years old sleep on average 7.2 hours, and according to the National Survey of Health (2017), 63.2% suffer from any sleep disorder. Hence, a sleep-deprived Chilean population may be more likely impacted by these transitions if the quality and amount of sleep are low and bad in quality.

[5] A detailed discussion of the mechanisms is available in the section 2, "Mechanisms".

administrative costs) but also indirect costs (loss of productivity) and these indirect costs are sometimes bigger than the direct costs. Moreover, automobile accidents lower the quality of employees' health and may result in loss of income and, in extreme cases, death. Around the world, the cost of automobile accidents represents between 3% and 5% of the GDP (Organization 2019). In Chile, car accidents have an associated cost of USD 5,888 million equivalent to 2.1% of the GDP.[6]

The identification strategy relies on the two usual approaches used in the literature. First, I exploit the sharp discontinuity created at the transitions where we set the clock forward (backward) by one hour at midnight in the spring (fall) using a Regression Discontinuity Design (RDD) to measure the immediate impact of the policy. Secondly, based on the variation in the date that DST starts and ends over the years, I use a DID approach to identify the effect of the policy over dates that are either DST or ST, depending on the year. The impact is identified only on those common dates in a total of two months after each transition. The estimates from the RDD show that the immediate net impact of the transition into DST and ST is zero. The DID approach supports this finding in the short-term. In the longer-term, using the DID method, I observe a net reduction in accidents on dates covered by DST in the spring season and a smaller reduction in the fall season. Conducting a heterogeneity analysis across geographical locations, rural/urban zones, and by driver's gender, I find that the total effect is mainly explained by accidents in the urban areas and the country's central zone, presumably since more than 75% of the accidents occur in these locations. I also calculate a cost-benefit analysis based on two approaches. These values should be considered a lower bound of the true saving costs. Implementing the DST regime on all the dates where I identify the effect would save between 108 and 162 million USD annually.

To disentangle the effect of the light and sleep disruption, I isolate the sleep disruption effect. To do so, I first look at the hours of the day unaffected by the change in the amount of light after both transitions. Secondly, I compare the number of automobile accidents within two weeks following the transition into DST (ST) over dates that are sometimes covered by the DST regime and sometimes by the ST regime. The two tests suggest sleep disruption occurs in the first few days following both transitions. Next, to isolate the light mechanism, I focus on dates two weeks onwards following both transitions that are not affected by the sleep disruption effect and on hours of the day that are impacted by the reallocation of light. I do not find conclusive evidence for the light mechanism when I analyze total accidents. I find a reduction in automobile accidents in the evening and morning under the DST regime and an increase in the evening under the ST regime. Therefore, the evening result supports the light effect; nevertheless, the morning result goes opposite to what it is expected. To address this issue, I present suggestive evidence that the under-reporting of minor accidents over the DST regime in the morning may be the reason that prevents from finding evidence for the light channel for total car crashes. Focusing on the morning, an analysis of only minor accidents replicates the unexpected signs. In contrast, an analysis for serious+fatal accidents, that are less likely to be not reported, lessens the bias and shows the expected signs

[6]To put this value in context, this value is close to the public expenditure on primary to post-secondary non-tertiary education, 3% of the GDP according to OECD data.

supporting the light mechanism. Therefore, I present evidence that ambient light reduces serious and fatal accident risk. Finally, using an alternative approach developed by Bünnings and Schiele 2020, I present evidence that darkness increases automobile accidents, favouring the light channel.

This research contributes to the literature studying the DST's impact on automobile accidents as follows. First, as I previously stated, I isolate the light and sleep disruption mechanisms following Smith 2016, which offers the most rigorous study of the effect of DST on fatal automobile accidents so far. I expand the analysis to total automobile accidents[7] since considering less severe automobile accidents[8] is relevant because they can be thought of as a more direct measure of labor productivity than fatal automobile accidents[9].

Second, I leverage on a longer time horizon of four months to investigate the longer-run consequences of DST on automobile accidents than the current literature[10]. I use a time window of two months after each transition that provides the following advantages over the current literature. On the one hand, it allows to separate and analyze the short and longer-run effects after both transitions and explore them far from the first month. On the other hand, it allows to test the sleep disruption impact following both transitions and to present evidence in favor of this mechanism following the transition into ST[11] in the same line as Jin and Ziebarth 2020 who find health effects following the fall transition. Additionally, by exploiting the heterogeneity in the multiple DST transition dates in Chile and a long time series of 17 years, I can control for month/date fixed effects, thus relaxing the concern that the findings are driven by a calendar effect and thereby precisely estimating the impact of the DST transitions.

Third, most papers on the DST regime's effects on automobile accidents have focused on developed countries and latitudes closer to the poles. To the best of my knowledge, only Goodwin et al. 2023 focus on a developing country such as Mexico. This paper adds evidence on a developing country in the southern part of the globe, spanning latitudes not previously studied for medium- to high-income countries. This is relevant since this paper provides evidence in an environment where

[7]Smith 2016 focuses only on fatal automobile accidents since the author argues that "many states do not maintain a uniform database of these less severe crashes". Focusing on fatal automobile accidents can only be considered as a lower bound of the effect of DST on automobile crashes. In contrast, I focus on all reported automobile accidents since the Chilean Police have maintained a uniform database during the period studied.

[8]In the case of Chile, a serious injury is when a person loses at least 30 days of work. A minor is when there is a loss between none to 30 days of work.

[9]Each automobile accident, depending on the level of severity, can be linked to loss of days of work, loss of income, and physical or psychological damage that can affect productivity in the short and even long-run. For instance, Mello et al. 2009 calculated that in Brazil, 42% of the costs of automobile accidents corresponds to a loss in productivity. This study calculates the average cost of a traffic accident in Brazil is US$ 5,167 overall, US$ 1,919 in accidents with no victims, US$ 52,942 in accidents resulting in injury, and US$ 247,647 in accidents resulting in death. Moreover, for the US, Ebel et al. 2004 found that people who did not suffer any kind of injury on average lost 0.3 days of work, and people who suffered some kind of injury lost between 7.7 and 2,202.4 days of work.

[10]Among those papers exploiting changes in dates (time) or states (geographic), they do not have a long time horizon to estimate the impact. For example, Sood and Ghosh 2007 and Smith 2016, for the US, have three to four weeks of common support after the transition into DST and one week after the transition into ST. Therefore, they only calculate a short-term impact around these transitions. Recently, James 2022 for Australia has a one-time/one-month leveraging state-level variation.

[11]Smith 2016 presented suggestive evidence that the sleep disruption effect in the fall transition is non-existent; however, the literature has presented evidence against this claim. See sections 2.

the consequences can differ in several ways from developed economies. For instance, health services in developing countries are of lower quality than in their developed counterparts, and thereby, any negative shock of these transitions can be amplified in a developing country, raising direct and indirect costs.

It also contributes to the literature investigating the effect of sleep on productivity. Many authors have found that the disruption in the sleep patterns or sleep time affects how individuals process information (Kamstra et al. 2000), cognitive performance (Van Dongen et al. 2003), school performance (Heissel and Norris 2018, Carrell et al. 2011), absenteeism (Metlaine et al. 2005, labor market outcomes (Gibson and Shrader 2018, Gillingham 2014), workplace accidents (Barnes and Wagner 2009), heart attacks (Jin and Ziebarth 2020, Toro et al. 2015) or cyberloafing (Wagner et al. 2012). I complement this literature by exploring the impact of losing and gaining one hour after both transitions in the same spirit than Smith 2016 and, unlike him, finding a significant effect when one hour is gained in the fall transition similar to Jin and Ziebarth 2020. Although I am not able to test changes in sleep time[12], the literature studying the impact of DST on sleep patterns has found suggestive evidence of a reduction in sleep duration after the spring transition (Barnes and Wagner 2009, Michelson 2011, Valdez et al. 1991, Depalo 2023) and extension after the fall transition (Jin and Ziebarth 2020, Michelson 2011, Depalo 2023).

The rest of the paper is organized as follows: section 2 describes in extensive detail the two main mechanisms affecting automobile accidents after every transition and describes previous literature on DST. Section 3 briefly gives background about the DST policy in Chile and the costs associated with automobile accidents. Section 4presents the data. Section 5 shows the two empirical strategies: RDD and the DID Day-of-Year FE approach. Section 6 presents the short- and longer-term results, discusses the mechanisms and shows the heterogeneity analysis. Section 7 calculates cost-benefit analysis and provide robustness checks to the main results. Finally, section 8 concludes.

2 Mechanisms

The exogenous reallocation of one hour of sunlight from the morning to the evening during DST and vice versa during ST affects automobile accidents through two mechanisms: altering the circadian rhythm and reallocating light. The first mechanism is the disruption of the circadian rhythm, defined roughly as a 24-hour cycle in the physiological processes of living beings. Any alteration of the circadian rhythm disrupts the amount and quality of sleep (Barnes and Wagner 2009)[13]. For example, when sleep quantity is lowered, there are negative effects on temporal memory, divergent thinking tasks, and attention (Harrison and Horne 2000, Ilies et al. 2007). Several studies have also concluded that less time sleeping increases the number of workplace accidents, train accidents, and driver-at-fault accidents (Transportation and Road 2009, Barnes and Wagner 2009, Legree et al.

[12]Similar to Smith 2016 who does not have information about the sleep time of the US population but he argues that these effects should come from this mechanism based on other studies.

[13]An example of disruption in the circadian rhythm is the ones that happen when flying across multiple time zones, commonly known as jet lag.

2003).

In the context of the DST transitions, entering (leaving) the DST regime causes clocks to be set forward (backward) by one hour at midnight. These transitions cause a shock in the circadian rhythm following both transitions. Harrison 2013b suggests that these transitions increase sleep fragmentation and sleep latency, cumulatively affecting duration and quality of sleep. This impact is shown to last between a few days and two weeks (Valdez et al. 1991, Vidaček et al. 1986). Furthermore, Barnes and Wagner 2009 indicate that the negative effect of gaining and losing one hour is asymmetric, with people adjusting more easily to delay than advanced phases (Folkard and Barton 1993, Monk et al. 2000). In the case of entering to the DST regime, one hour is lost. I would expect that people adjust by going to bed earlier. However, there exists evidence of sleep reduction between 30 and 40 minutes, sleep fragmentation, and sleep latency (Barnes and Wagner 2009, Michelson 2011, Valdez et al. 1991, Depalo 2023). In this case, these effects altogether produce fatigue and drowsiness. Moreover, Costa et al. 2022 show that the transition into DST deteriorates sleep and increases time stress, affecting physical and emotional health. Following the transition into ST, there also exists evidence of extension of people's sleep duration between 11 and 40 minutes (Jin and Ziebarth 2020, Michelson 2011, Depalo 2023) or no change (Barnes and Wagner 2009), and also of sleep fragmentation (Harrison 2013a, Meyer 2004, Violani et al. 2002). In this context, sleep fragmentation increases fatigue and drowsiness, and an increase in sleeping duration reduces them, so the net effect is unclear.

The second channel is the reallocation of light from the morning (evening) to the evening (morning) under DST (ST), and at the same time, all other activities, including commuting stay the same. Environments, where more daylight is available, suffer from fewer automobile accidents (Sood and Ghosh 2007 and Bünnings and Schiele 2020). The opposite is also true: in the absence of light, more automobile accidents occur[14].

Both channels explained above may affect the likelihood of automobile accidents. First, following the transition into the DST regime in both the short and longer-run, the reallocation of light from the morning to the evening creates a safer (riskier) environment in the evening (morning)[15]. The reverse holds after the transition into the ST regime. A daily net reduction of automobile accidents would be expected if the flow of automobiles differs in the morning from the evening and the additional hour of light is allocated to the higher automobile flow time frame while the commuting schedule stays the same. Evidence in favor of a net reduction of automobile crashes by the light mechanism during the DST regime is found by J. M. Sullivan and Flannagan 2002, Sood and Ghosh 2007, Laliotis et al. 2023 and Bünnings and Schiele 2020. However, Smith 2016 found no net effect for the US despite light playing an important role in the morning and evening. In the context of Chile, according to the fifth Census of Traffic Flow from the Engineering and Construction Management Department of the Universidad Católica de Chile (DICTUC 2006), the flow of automobiles is heavier in the evening than in the morning over the year. Therefore, if the

[14]Also, it has been shown that additional ambient light plays an important role in reducing crime (Doleac and Sanders 2015, Domınguez and Asahi 2023 Toro et al. 2019, Tealde 2021)

[15]There exists evidence that darkness increases the number of automobile accidents (Bünnings and Schiele 2020).

government can set the DST regime for the whole year and indeed, more automobile accidents occur in the evening, the policy would be cost-effective, reducing the annual number of automobile accidents. Some authors in favor of the DST regime have calculated that imposing DST all year would reduce the net amount of automobile accidents by 180 fatal collisions for the US (Ferguson et al. 1995) and 25 for the UK (Bünnings and Schiele 2020).

The second active mechanism is the disruption of sleep patterns in the two weeks following both transitions. After the spring transition, sleep fragmentation, sleep latency, and a reduction in time spent sleeping reduce alertness and concentration, making people more prone to automobile accidents. The evidence is mixed, with some authors finding an increase in automobile accidents (Smith 2016, Bünnings and Schiele 2020, Varughese and Allen 2001, Goodwin et al. 2023), while others found no effect (Sood and Ghosh 2007, Lahti et al. 2010, Lambe and Cummings 2000, James 2022). After the fall transition, sleep fragmentation induces sleep of worse quality but the amount of sleep may increase, therefore, the net effect is uncertain. Authors also have found mixed results, with some finding a reduction in automobile accidents (Coren 1996, Sarma and Carey 2015, Askenasy et al. 1997), others no effect (Lahti et al. 2010, Lambe and Cummings 2000) while others do find an increase (Goodwin et al. 2023, Hicks et al. 1983). More recently, Depalo 2023 found evidence that the transition into DST increased accidents at work while the transition out of DST reduced them.

Table 1 summarizes the expected impacts of both mechanisms on automobile accidents.

Table 1: Expected Effect on Automobile Accidents

	DST regime		ST regime	
	Short Run	**Long Run**	**Short Run**	**Long Run**
Morning	(+)	(+)	(?)	(-)
Evening	(?)	(-)	(?)	(+)
Least Light Impacted Hours	(+)	(0)	(?)	(0)

Note: The overall expected signs of the effects of the transitions and time regimes on automobile accidents. A detailed explanation is on the appendix in section A

3 Daylight Saving Time in Chile

In 1970, the Chilean government adopted the DST policy between October and March, starting the second Sunday of October of every year at midnight to save energy. Between 2008 and 2016, Chile changed the transition dates eight times. To the best of my knowledge, it is one of the countries that has changed the DST transition dates the most.

The reasons for these continuous changes have been very different: in 2008, it was due to a drought; in 2010, an earthquake; and in the following years, it was due to reducing energy consumption. In 2015, Chile was on the DST regime for the entire year. In 2016, the government,

based on expert opinions in different fields and public opinion, decided to implement again these transitions[16]. Table 2 shows a summary of the policy changes. I use these continuous changes in the policy as a relevant source of variation to identify the effect of DST on automobile accidents in the short- and longer-run. Also, this strategy reduces any concern that the effects are correlated with calendar effects. Additionally, the information from above shows that the multiple DST transition date changes are unlikely to be correlated to automobile accidents.

Table 2: DST Dates Between 2002-2018

Period	DST off	DST on
2002-07	2nd Sunday in March	2nd Sunday in October
2008	Last Sunday in March	2nd Sunday in October
2009	2nd Sunday in March	2nd Sunday in October
2010	First Sunday in April	2nd Sunday in October
2011	2nd Sunday in April	3rd Sunday in August
2012	Last Sunday in April	2nd Sunday in September
2013	Last Sunday in April	2nd Sunday in September
2014	Last Sunday in April	First Sunday in September
2015	DST	DST
2016-18	2nd Sunday in May	2nd Sunday in August

Source: Decree-Law 1489: Regulation of the official time changes. Ministry of the Interior, Chile.

Finally, it is important to highlight that Chile is located in the southern hemisphere, and as such, the respective spring and fall months are reversed compared to countries in the northern hemisphere. In Chile, the transition into DST occurs between August and October (spring) and the transition into ST between March and May (fall). Also, Chile is one of the longest countries in the world (4,300 km), and due to having a one-time zone (GMT-4), when the DST transition occurs, the difference between sunrise and sunset differ between municipalities[17].

4 Data

I use data from administrative records of CONASET (National Commission of Transit Security from its Spanish acronym), which obtains data from the police. This data contains daily individual records of each reported automobile crash between 2002 and 2018 by municipality[18]. Information on accidents includes the exact time of the accident, municipality, zone (urban/rural), type of accident, cause, the number of people involved in the accident, their age, gender of those involved, and injury level.

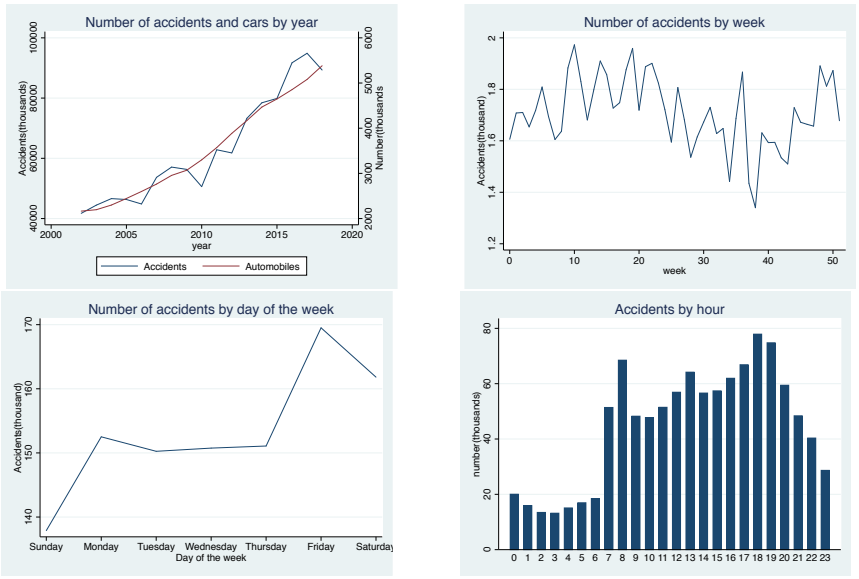
[16] One of the main reasons was due to the increase in student absenteeism during the winter when sunrise occurs at later hours (PNUD 2016).

[17] In 2017, the southernmost region in Chile, Magallanes (closest to the Antarctic), begun to use the DST regime during the entire year to have more light in the evening during the winter.

[18] Municipality is the smallest administrative division of local government. It is also known as a municipality and is most similar to a county in the United States.

In Chile, both the number of transit accidents and the number of automobiles have been increasing over the years[19]. For this reason, a policy focused on decreasing the number of accidents is relevant since the costs associated with crashes are increasing over time (Figure 1). Also, both the number of accidents and traffic tends to be higher in the afternoon, especially in the evening hours (Figure 1 and DICTUC 2006). Hence, if light plays an important role in road accidents, a policy that reallocates light from the morning to the evening would be cost-effective. Additionally, figure 1 displays that accidents are more likely to occur on the weekend.

Figure 1: Distribution of Accidents by Year, Week, Day of the Week and Hour



Note: Own calculation. Illustration based on CONASET data.

Since Chile is an extraordinarily long country, it is important to control for heterogeneity at sunrise and sunset (Smith 2016). Using the longitude and latitude of the municipality where the accident occurred, I calculate the exact time of sunrise and sunset using the algorithm of Meeus 1991 coded in Stata by Gibson and Shrader 2018. The exact days of the transitions into DST (ST) are taken from Decree-Law 1489 of the Government of Chile.

The dependent variable is the natural log of accidents each day at the national level. As in previous papers, the main reason for doing this is the scarcity of automobile accidents daily at the municipality level (see table 3). Aggregating the number of accidents allows me to gain statistical power and smooth out potential confounders such as weather, which could drive the results in some municipalities or regions but likely not the entire country. Additional analysis is done to evaluate heterogeneous effects by geographical locations, urban/rural zones, and using different specifications at a more granular level, and they are shown in the appendix.

[19]The number of automobile accidents increased from 144 in 2002 to 244 in 2018 on average. A possible explanation for this fact is that the number of automobiles has been increasing over the years from around 2 million in 2002 to 5 million in 2018, despite an increase in population of only 2.5 million (Figure 1, Panel A).

Table 3: Descriptive Statistics for Accident Variables.

	Mean	Std. Dev.	Min	Max	Obs
	(1)	(2)	(3)	(4)	(5)
Total accidents	173.45	54.83	37	424	6,209
Fatal accidents	3.96	2.28	0	16	6,209
Serious accidents	16.67	5.18	0	42	6,209
Minor accidents	152.81	52.83	37	390	6,209

Source: CONASET data. Note: Daily averages for the period between January 1st 2002 and December 31st 2018.

5 Empirical Strategy

5.1 Regression Discontinuity Design

To estimate the effect of the transitions into DST and ST, I exploit the discrete change in automobile accidents the day before and the day after the transitions using a Regression Discontinuity Design (RDD). To identify this effect, I rely on a natural experiment experienced by the population where the clock is set forward (backward) by one hour in the spring (fall), exploiting within-year variation. This generates an exogenous reallocation of sunlight from the morning to the evening in the spring and vice versa in the fall. Comparing automobile accidents just before and after the transition provides an estimate of the impact of the DST policy.

To eliminate any seasonal effect, in a first step, I demean the log of car accidents by day-of-week[20], month and year. Day-of-week controls for the fact that automobile accidents are unequally distributed over the days; for instance, they are higher on Fridays; month controls for the fact that car accidents are seasonal through the year[21], and year controls for long-run time trends. Next, in the second stage, I use the usual RDD specification with the demeaned log of automobile accidents as the dependent variable. I estimate the following equation:

$$\begin{aligned} Ln\ Accidents_{dy} =& \beta_0 + \beta_1 DST_{dy} + g(DaystoTran_{dy}) \\ &+ g(DST_{dy} \times DaystoTran_{dy}) + \varepsilon_{dy} \end{aligned} \tag{1}$$

$Ln\ Accidents_{dy}$ is the log of automobile accidents at day d and year y, DST_{dy} is an indicator equal to one if day d in year y falls under Daylight Saving Time and $DaystoTran_{dy}$ is the running variable, measuring time in days before and after the DST transitions and centered at the transition date in each year. $g()$ is a function of the running variable.

To estimate a consistent parameter of interest, β_1, the treated and untreated rate of automobile accidents must vary continuously with a date around the transition. In other words, if all other

[20] Also, I perform this estimation without the first day after the DST transition (Sunday) to eliminate any suspicion that the results are driven by the variation of accidents on this day. Results are shown in table D.10.

[21] Following the same specification in Smith 2016, I also estimate the RDD approach without month FE in table D.15. The results remain the same.

factors affecting accidents, aside from DST, are continuous around the transition date, then the RDD provides a consistent estimate of the effect of the DST transition.

In the main specification, I use a local linear regression, $g()$, since it performs better in RDD settings than higher-order polynomials of the forcing variable (Gelman and Imbens 2019). The baseline specification uses the optimal bandwidth of the Calonico et al. 2017 to choose the number of days at each side of the cut-off and a triangular kernel. As a robustness check, in the appendix, I present the estimates using the two bandwidth selectors that choose a different bandwidth at either side of the cutoff[22].

5.2 Difference-in-Difference: Day of Year Fixed Effect

RDD provides an estimate of the immediate effect of entering the DST and ST regimes on automobile accidents around the transition date, however, it does not allow us to infer the effects of the DST policy in the longer-run (Smith 2016). To this end, I leverage on the variation of the coverage of the DST regime, in other words, on the multiple dates that DST starts and ends over the years, exploiting differences between years. This variation allows me to control for seasonality and provides more common support and heterogeneity for the treatment variable. These changes create some dates that are DST in some years and ST in others. For example, the earliest transition date into DST is the second Sunday in August, and the latest is the second Sunday of October. Thus, I have two months of common support between years to identify the effect of the DST regime on automobile accidents. Similarly, the earliest transition into ST starts on the second Sunday of March, and the latest one, on the second Sunday of May. Again, I have two months of common support to identify the effect of the DST regime over fall dates. Therefore, unlike Smith 2016 and Sood and Ghosh 2007, this study has longer common support of four months to estimate the effect of the DST regime on automobile accidents. Besides, these two months of common support in the fall allow me to measure the effect of sleep disruption around the transition into ST. Smith 2016 provides suggestive evidence that the sleep disruption effect in the fall transition is non-existent; however, the literature has presented evidence against this claim[23](Jin and Ziebarth 2020).

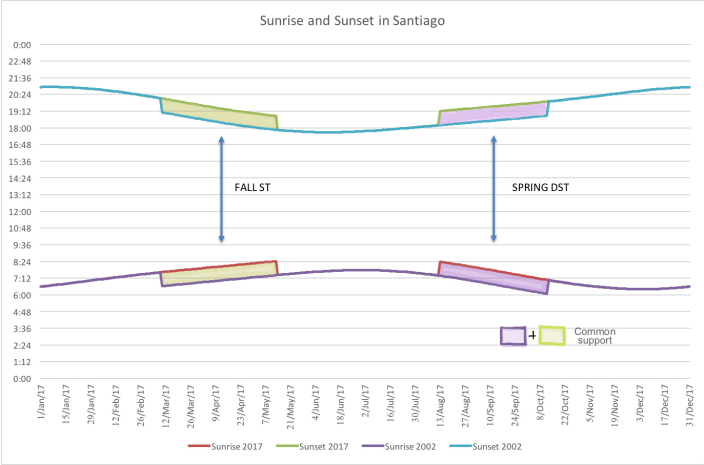
Under the assumption of common trends, meaning that in the absence of DST, the trends of automobile accidents would have trended in parallel between treatment and control dates and that the policy decision was exogenous to the dependent variable, I build a natural counterfactual for the treatment group determined by those years where the DST regime was implemented as the law regularly established it. Figure 2 shows the sunrise and sunset hours in 2002 and 2017 in Santiago and the dates of the transitions reflecting that the same day/month (but different years) had a different exposure to sunlight at specific times. For example, on March 12, 2002, the sunrise was at 6:40, while on March 12, 2017, the sunrise was at 7:40. In other words, between 6:40 and 7:40, the amount of ambient light was lower in 2017 than in 2002. Also, the civil twilight starts (ends)

[22]For robustness, I also estimate the RDD using an epanechnikov and a uniform kernel, and higher polynomial orders, results remain the same. Results are shown in tables D.10 and D.11

[23]For a detailed discussion, see section 2

on average 30 minutes before (after) sunrise(sunset), thus affecting light conditions[24].

Figure 2: Change in the Amount of Light Caused by the DST Transitions



Note: The sunset and sunrise times are for the capital city, Santiago. This figure is based on Figure 1 from Kotchen and Grant 2011 and from Smith 2016. Soft green (soft lila) area corresponds to the dates covered by the fall ST (spring DST) or common support to identify the effect of ST (DST) on automobile accidents. Source: Author's calculations.

Taking advantage of the exogenous variation of the policy, I estimate a Difference-in-Difference model with Day of Year Fixed Effect:

$$\begin{aligned} Ln\ Accidents_{dy} = \beta_0 + \beta_1 SpringDST_{dy} + \beta_2 FallST_{dy} + DayofYear_d \\ + DayofWeek_{dy} + Year_y + Holiday_{dy} + \epsilon_{dy} \end{aligned} \tag{2}$$

$LnAccidents_{dy}$ is the log of the number of automobile accidents at day d and year y, $SpringDST_{dy}$ is a dummy variable equal to one if day d in year y is under the DST regime in spring, $FallST_{dy}$ is a dummy variable equal to one if day d in year y is under the ST regime in fall[25] $DayofYear_d$[26] is the specific day of the year, $DayofWeek_d$ is the day of the week, $Year_y$ is the year, $Holiday_{dy}$ is a dummy equal to one if the day d in year y is a holiday. β_1 and β_2 are the coefficients of interest and are interpreted as the average effect of the DST and ST regime on automobile accidents over the different dates in the spring and fall seasons, respectively. Notice that this specification is a usual DID model since the variables $SpringDST$ and $FallST$ are effectively interaction terms between

[24]In particular, the years 2002 and 2017 were chosen to construct this figure because, in 2002, the DST regime lasted the least and in 2017 the most.

[25]To interpret the variable *FallST* during the DST regime, it must be done by adding a negative sign to the effect. For instance, if the variable *FallST* is equal to 0.05, it means that during the ST regime, car accidents increase by 5%. During the same period, but under the DST regime, car accidents decrease by 5%.

[26]Following Smith 2016, I create dummies corresponding to each unique month-day pair. In this way, leap years are matched exactly with their corresponding date. This helps to control for any effect on particular days (e.g., September 17th is the day before national holidays, and although this is not a statutory holiday, many individuals work a half-day and travel by automobile to their hometown to celebrate).

the day of the year and the year-specific of the DST regime. Both effects are identified in the common support: August 12th-October 15th for the spring DST, and March 9th-May 15th for the fall DST. Figure 2 illustrates switching dates or the common support where both effects of DST and ST are identified.

This DID approach calculates a different effect than the RDD approach. RDD estimates the effect of entering DST (ST) right after the transition so that it can be thought of as the instantaneous effect of DST on automobile accidents. Under the DID specification, β_1 measures the average effect of the DST regime on automobile accidents over the two months that some years are DST and others ST during the spring. Because of the day-of-year fixed effects, only days that experience a change in DST status over the period covered by the data contribute to estimating this effect. Therefore, The DID model estimates the effect of not only the initial difference in automobile accidents due to the DST transition but also the effect that persists for several weeks. Importantly, neither approach can tell us much about DST's effect on dates far from DST transitions, such as mid-summer. Similarly, β_2 measures the average effect of ST over those two months that some years are ST and others DST in the fall switching dates.

6 Results

6.1 Regression Discontinuity Design

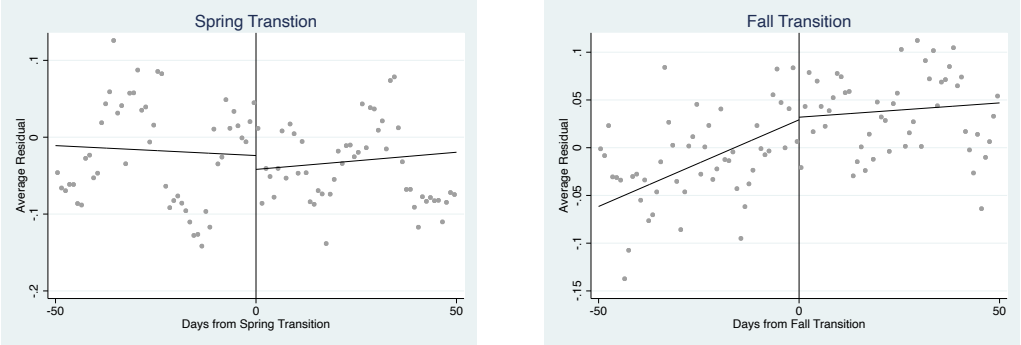
Figure 3 plots the average residuals from a regression of the natural log of total accidents on day-of-week, year dummies, and holidays. The x-axis represents days from spring and fall transitions, with zero representing the transition date. In panel A, I observe a subtle drop in accidents after the transition into DST (spring transition). Panel B does not show a significant discontinuity after the transition out of DST (fall transition). Also, I split the figures into types of accidents. Figures E.1 and E.2 also show no effect around the transitions for minor, serious and fatal accidents[27].

Table 4 complements the previous analysis and illustrates the effect of the transition into and out of DST on road accidents from figure 3. Column 1 shows that the transition into DST is associated with no significant impact on the number of accidents[28]. In column 3, I also find no significant effect of the transition into ST on automobile accidents.

[27] A detailed set of figures of the RDD estimates by transition/type of accidents/hour of the day can be found in the appendix.

[28] To address any concern that the first Sunday after the spring transition is one hour shorter, I adjust the number of car accidents. Following Smith 2016, for the missing hour, I multiply by 2 the number of accidents occurring between 1:00 and 2:00 a.m. and divide by 2 the number of accidents between 11:00 p.m. and 0:00 a.m. in the fall transition. To discard that this way of adjustment is the reason for the results, first, I run the same regression without adjustment; secondly, I eliminate the transition day of the estimation; and thirdly, I multiply the number of accidents for the first transition into DST by 24/23 and the day before transitioning out DST by 24/25. Regardless of the adjustment, results remain identical. Results are available in tables D.10 and D.12 for all and the least light-impacted hours. Also, tables D.11 and D.13 show alternative polynomials for the RDD estimates. All results are consistently robust to different specifications.

Figure 3: Residual Plots from the Transitions into DST (spring transition) and into ST (fall transition)



The residuals are computed from a regression of natural log of automobile accidents on day-of-week, year dummies and holidays. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Table 4: RD Estimates of the Impact of Entering DST and ST on Automobile Accidents.

	Spring DST		Fall ST	
	(1)	(2)	(3)	(4)
Transition	-0.0059	-0.0046	0.0059	0.0004
	(0.0120)	(0.0121)	(0.0130)	(0.0159)
Bandwidth Selector	mse	msetwo	mse	msetwo
Bandwidth Left	81	76	65	38
Bandwidth Right	81	84	65	47
Observations	2610	2581	2031	1338

Note: The dependent variable is the log of automobile accidents. All specifications control for day-of-week, holiday, month and year, and they use a first-order polynomial. DST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

6.2 Difference-in-Difference: Fixed Effect Model

Table 5 shows the results for DID the Day of Year fixed-effect model. This identification strategy aims to capture the longer-term effects and also serves as a robustness check of the short-term effects of the RDD. The first column examines the effect of the spring DST and the fall ST, while the fourth column divides the spring DST and fall ST into the first and second week of DST or ST and the remainder of the spring DST and fall ST. The second and third columns test for pre-trends using either the variable one or two weeks before each transition, respectively. Finally, I use national daily prices for gasoline to control driving patterns since fuel prices significantly influence Vehicle Miles Traveled (VMT) (Gillingham 2014).

Column 1 of Table 5 shows that over the two months of common support under the spring

DST, there is a significant reduction of 2.7% in automobile accidents. Fall ST is associated with a 1.1% increase in accidents, albeit it is not precisely estimated (p-value of 13%). In column 4, I split these results by testing a short-term and longer-term impact. I find a negative effect of 1.5% and 1.9% for the first and second weeks following the transition into DST, respectively, but both are not significant. The remainder of the spring DST is associated with a 3.6% significant reduction in automobile accidents. Moving to the transition into ST, I find a zero effect in the first week and a 2.4% increase for the second week following the transition. The remainder of the fall ST is associated with a 1.6% significant increase in automobile accidents. The results in the first two weeks align with what I found in the RDD: no significant impact around the transition into DST and ST. Finally, columns 2 and 3 provide an "event study" type of evidence as to whether there seem to be "pre-trends" on the days leading into and out of DST. No significant effect is found one or two weeks before entering DST or ST.

6.3 Mechanisms

As a result of the zero net effects at both transitions, could it be that different effects over the day are cancelling out each other, or does no impact exist? To address this question and to test the sleep disruption and reallocation of the light mechanisms, I analyze different portions of the day: the morning, the evening, and the rest of the day (the least light-impacted hours). It is important to highlight that in the morning and evening hours, there exists a change in the amount of light due to the reallocation of it. Therefore, both mechanisms, light reallocation and sleep disruption, should be present, at least in the short-term. In the rest of the day, the amount of light is constant. Therefore, I would expect circadian rhythm alteration (sleep disruption) to be the main factor driving any change in automobile accidents during these hours.

To do this analysis and, given the length of the country, it is necessary to control for heterogeneity in the sunrise and sunset in each location[29]. Then, I define the average day of transition within the common support of the transition into DST and ST for each municipality[30]. Using the average day of transition, I define the *Morning*, *Evening*, and *Least Light Impacted Hours* classifications. To determine the *Morning* and the *Evening*, I pick +/- three hours[31] around the average hour of the sunrise and the sunset between the day before and after the transition, respectively[32]. The *Least Light Impacted Hours* consists of the 12 hours left that cannot be placed into the morning or

[29] I use the algorithm of Meeus 1991 which links the longitude and latitude of the municipality where the accident took place with the hour of sunrise and sunset. This procedure allows me to control for these differences, keeping the estimations at the country level.

[30] These correspond to April 4th in the case of the transition out of DST and to September 17th in the case of the transition into DST.

[31] Also, I pick +/- two, two and a half, thee and half and four hours around the average hour of the sunrise and the sunset between the day before and after the transition respectively, and the results remain the same. Results are available upon request.

[32] For example, sunrise in Arica (the northernmost city) is at 6:34 AM before the transition into DST and at 7:34 AM after the transition date, so morning for Arica will be between 4.04 AM and 10:04 AM. Sunrise in Santiago (the capital city in the middle of the country) is at 6:38 AM before the transition and 7:38 AM after the transition date, so the morning for Santiago will be between 4.08 AM and 10:08 AM .

Table 5: FE Estimates of the Effect of DST and ST on Automobile Accidents by different time horizons.

	(1)	(2)	(3)	(4)
Spring DST	-0.0270***			
	(0.00855)			
First week of DST		-0.0159	-0.0168	-0.0148
		(0.0146)	(0.0147)	(0.0145)
Second week of DST		-0.0200	-0.0203	-0.0187
		(0.0138)	(0.0138)	(0.0136)
Remainder of Spring DST		-0.0349***	-0.0357***	-0.0331***
		(0.00972)	(0.00992)	(0.00927)
Fall ST	0.0118			
	(0.00826)			
First week of ST		-0.0051	-0.0014	-0.0037
		(0.0136)	(0.0137)	(0.0135)
Second week of ST		0.0220*	0.0260**	0.0237*
		(0.0124)	(0.0125)	(0.0122)
Remainder of Fall ST		0.0135	0.0184*	0.0152*
		(0.00976)	(0.00996)	(0.00836)
One week before DST			-0.00860	
			(0.0138)	
One week before ST			0.0129	
			(0.0120)	
Two weeks before DST		-0.0107		
		(0.0131)		
Two weeks before ST		-0.0110		
		(0.0125)		
Observations	6,209	6,209	6,209	6,209
$Adj - R^2$	0.845	0.845	0.845	0.845

Note: The dependent variable is the log of automobile accidents; all regressions use day-of-year, day-of-week, holiday, and year dummies. The remainder of spring DST (fall ST) is a dummy variable equal to one if the day belongs to dates following the first two weeks of DST (ST) and after (before) July 1st. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

evening.

I start the analysis using the RDD method for the transition into DST in Panel A. I begin by analyzing the least light-impacted hours where only the sleep disruption effect is present. Column 5 of Panel A in table 6 shows a 6% increase in accidents right after the spring transition. This finding is in line with the existence of the sleep deprivation channel. In the morning, column 1 displays a significant and surprising reduction of 7.5%. No significant effect is found in the evening. In the next section, I discuss this unexpected sign in the morning.

Moving to the transition into ST in Panel B, for the least impacted hours by light, in column 5, I find a 3.9% significant reduction in automobile accidents, supporting the hypothesis of the sleep disruption channel. In this case, it seems that more time sleeping prevails over sleep fragmentation, as I discussed in section 2. I acknowledge that I do not have the data to test changes in sleep time;

however, some papers have found that this is true for both transitions (Jin and Ziebarth 2020, Michelson 2011, Depalo 2023, Barnes and Wagner 2009). Jumping into the morning, in column 1, I find an increase in automobile accidents contrary to my expectations, similar to my reversed results from the morning in the spring transition. Finally, no effect is found in the evening analysis in column 3[33].

To show the validity of the estimates in table 6, I perform a falsification test based on permutation tests. To do this, I assign fake transition dates 120 times starting the following date out of the optimal bandwidth for all significant coefficients. Figure E.11 in the appendix presents the distribution of these parameters where the vertical lines correspond to the baseline estimates. The morning results are associated with implied p-values of 9% and 12%, and the coefficients from the least light-impacted hours with 5% and 6% for the transition into DST and ST, respectively. These results ensure the validity of the main RDD results at conventional levels.[34]

Next, in table 7, I explore the DST/ST impact on different portions of the day as I defined previously using the DID model. Column 1 presents the estimates for all hours as the baseline. I begin by testing the least light-impacted hours as an additional test for the sleep disruption mechanism. Column 4 indicates that the first week following the transition into DST is associated with a significant increase of 3.9% in automobile accidents. The following week, the number of accidents reduces by 2.3%, although it is not precisely estimated. For the remainder of DST, the effect drops to almost zero. Also, column 4 tests the effect of sleep disruption after the transition into ST. During the first week, the effect equals a reduction of 2.8% in automobile accidents. The remaining time horizons are not significantly impacted[35]. These results support the hypothesis that the transition into DST and ST generates a short-term sleep shock in line with the results from the RDD, and this effect fades away throughout the first two weeks in line with the Valdez et al. 1991 findings. The absence of the effect in the second week for the transition into ST, unlike the second week of the DST transition, is supported by the hypothesis of the asymmetric effect proposed by Barnes and Wagner 2009. They argue that it is more difficult to adjust to the spring than the fall transition. Recently, Depalo 2023 in the context of accidents at work, found evidence of a rise of accidents following the spring transition into DST and a reduction following the fall transition into ST, presumably explained by the disruption of the sleep patterns given the nature of the accidents at work[36].

A compelling feature of the DID approach is that it allows to isolate the light mechanism by

[33] In the appendix, from figures E.3 to E.8, I show the RDD plots after both transition for different times of the day and type of accident.

[34] In the appendix, I also present the same permutation tests for estimates for all types of accidents as long as they are statistically significant. Overall, from figures E.12 to E.15, they support the validity of the disaggregated estimates.

[35] Table D.14 shows the DID results without adjusting the by-hour missed in spring and the additional hour in fall and without the transition date. The results are similar, although the coefficient associated with the first week of DST is smaller but statistically not different from the main estimated effect.

[36] Although he is not able to separate the light and sleep mechanisms since he does not have the exact time of the accidents, it is more likely that accidents at work after these transitions are related to changes in concentration and drowsiness which impact work productivity

Table 6: Regression Discontinuity estimates by different portions of the day

	Morning		Evening		Least Light Impacted Hours	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Spring DST						
Transtition	-0.0752***	-0.0864***	-0.0148	-0.0161	0.0601***	0.0612***
	(0.0269)	(0.0266)	(0.0157)	(0.0154)	(0.0194)	(0.0177)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	48	40	74	76	45	50
Bandwidth Right	48	68	74	80	45	64
Observations	1552	1744	2384	2513	1456	1840
Panel B: Fall ST						
Transition	0.0538***	0.0582**	0.0037	0.0057	-0.0397**	-0.0378*
	(0.0204)	(0.0228)	(0.0202)	(0.0207)	(0.0204)	(0.0204)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	71	60	46	41	44	38
Bandwidth Right	71	51	46	49	44	53
Observations	2031	1338	2285	1792	1488	1456

Note: The dependent variable is the log of automobile accidents. All specifications control for day-of-week, holiday, month and year, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in automobile accidents that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in each municipality. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

focusing on days and hours unlikely to be affected by the sleep disruption effect, the remainder of the DST/ST. Column 2 shows that in the morning, there exists a significant reduction of 6% in automobile accidents over the remainder of the spring. During the remainder of the fall ST, no effect is found. Column 3 shows that in the evening, in the remainder of the spring DST, there is a significant reduction of 5.7% . For the transition into ST, the remainder of the fall ST shows an increase of 4.3%. Based on the results from above, I do not find conclusive evidence for the light mechanism. The evening results support ambient light being an important mechanism explaining the reduction of automobile crashes, but the morning results do not. In the next section, I explore a possible reason of why I do not find evidence of the light channel.

Finally, I also run the same specification from above, adding the coefficients for one or two weeks before entering DST and ST in an "event study" exercise. Table D.2 shows that none of these coefficients associated were significant for any portion of the day, indicating suggestive evidence of the absence of pre-trends.

Table 7: Difference-in-Difference estimates by different portions of the day

	All Hours	Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)
First week of DST	-0.0148	-0.0665**	-0.0388**	0.0394**
	(0.0145)	(0.0276)	(0.0191)	(0.0186)
Second week of DST	-0.0187	-0.0607**	-0.0325*	0.0244
	(0.0136)	(0.0256)	(0.0174)	(0.0178)
Remainder of Spring DST	-0.0331***	-0.0619***	-0.0569***	0.0074
	(0.00927)	(0.0172)	(0.0128)	(0.0117)
First week of ST	-0.00370	0.0269	0.00254	-0.0284
	(0.0135)	(0.0238)	(0.0180)	(0.0183)
Second week of ST	0.0237*	0.0600***	0.0228	0.0040
	(0.0122)	(0.0223)	(0.0175)	(0.0174)
Remainder of Fall ST	0.0152*	0.0010	0.0426***	0.0057
	(0.00836)	(0.0174)	(0.0125)	(0.0128)
Observations	6,209	6,209	6,209	6,209
$Adj - R^2$	0.845	0.721	0.762	0.780

Note: The dependent variable is the log of automobile accidents; all regressions use day-of-year, day-of-week, holiday, and year dummies. The remainder of spring DST (fall ST) is a dummy variable equal to one if the day belongs to dates following the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

6.4 Possible Explanation for the Morning Results

One reason that can explain the unexpected results in the morning in the short-term for the RDD and DID, and the unexpected coefficient for the remainder of DST, is the under-reporting of minor

automobile accidents. Although there is no current evidence for under-reporting in Chile, research in different countries has documented that the lesser the severity of the automobile accident, the higher the probability of not reporting it, therefore increasing primarily the rate of miss-reporting of minor accidents (Singh et al. 2018, Janstrup et al. 2016, Dandona et al. 2008, Amoros et al. 2006, Alsop and Langley 2001). In this context, a plausible hypothesis that can explain the unexpected signs in the morning is that minor accidents are under-reported over the DST regime in the morning hours. If this were to occur, the resulting estimates would be biased towards a reduction in accidents immediately following the transition into DST and towards an increase right after the transition into ST. Additionally, over dates far from the transitions, the DID coefficient should be biased towards negative and positive signs in the DST and ST, respectively.

To provide evidence on this, I run an additional analysis where I divide my sample between minor and serious/fatal accidents[37]. The key assumption needed and supported by the literature is that serious and fatal accidents are less likely to be unreported. Therefore, focusing on these types of accidents, I should not find unexpected signs in the morning or at least lessen the negative effect in the DST or the positive or null in the ST. The tables for the following analyses are available in the appendix.

I begin by analyzing minor car accidents. Columns 3 and 4 in table D.16 show the unexpected signs in the morning using the RDD approach similar to the morning results from table D.17 using the DID method where I find a reduction of 7% after the DST transition and an increase of 7% after the ST transition. Total and minor accident results align since minor accidents correspond to 88% of all car accidents. In contrast, when I add up serious and fatal accidents, columns 3 and 4 in table D.18 show no significant effects right after both transitions in the morning, but in the case of the transition into ST, the correct sign[38]. Table D.19 supports the previous findings in the first two weeks of DST, and in the case of the second week, I find a positive and significant increase of 8%. In the case of the first two weeks of ST, I find coefficients with the correct negative signs but not significant.

More interesting is the analysis for the longer-term. For minor accidents, table D.17 shows that the remainder of the spring DST reduces accidents by 6%, and the remainder of the fall ST has no effect in the same line for the analysis for total accidents. In contrast, for serious+fatal accidents, table D.19 shows that the negative sign of the remainder of the spring DST disappears, finding a null impact, therefore lessening the downward bias seen in the analysis of the total accidents. The coefficient of the remainder of the fall ST becomes negative and significant, reducing serious and fatal accidents by a 10%, lessening the upward bias as how it was hypothesized and supporting the light mechanism. Moreover, the evening results for the remainder of spring DST and fall ST keep supporting the light channel for all types of accidents[39]. Therefore, I show evidence that ambient

[37] Adding up serious and fatal accidents helps me to get more statistical power than studying fatal crashes alone since they both together compose 12% of the accident sample compared with the 4% of fatal accidents

[38] In the appendix, in figures E.9 and E.10, I show the plots of the RDD after both transitions for different times of the day.

[39] An even more accurate test would be to use only fatal car accidents since they are for sure reported. The downside of using this accident category is that the average daily fatal accident is four, making all the estimates

light reduces serious and fatal accident risk. In conclusion, the evidence presented above provides suggestive evidence that the reason why the morning results are showing the wrong signs aligns with a situation where the under-reporting of minor accidents happens over the DST regime.

A story behind this hypothesis and these supporting results could be that drivers in the morning face the rush to get to work combined with a sudden darker environment. It has been shown that unpredictability of commuting times in morning hours increases stress levels and salivary cortisol (Evans et al. 2002) and that the absence of ambient light is associated with cognitive impairments, including slower cognitive processing as well as impaired working and spatial memory (O'Brien et al. 1993, B. Sullivan and Payne 2007, Shi and Chen 2023, Davis et al. 2017). Both mechanisms interact, leading drivers to mistakenly not report a minor accident, presumably since they might under-estimate the severity (for example, very minor accidents), and combined, if reported, with increased commuting time. Moreover, if there is a sleep disruption effect around the transition into DST, groggy drivers may be less likely to report a very minor accident following the transition.

As an additional exercise to test the light mechanism, I follow the methodology developed by Bünnings and Schiele 2020. The authors use variation in the offset and onset of darkness across time and space[40]. Table D.1 shows that the overall results are consistent with what Bünnings and Schiele 2020 found: darkness increases accidents by an average of 11% by hour. To investigate the unexpected morning result, I focus on hours of the morning when the same hour experiences changes in the amount of sunlight over the year. Columns 2 and 3 show that darkness increases all automobile accidents from 10 to 25% for all accidents and by each type/definition of accident[41]. The evening results also align with the hypothesis of the light mechanism. Consequently, I provide additional suggestive evidence that darkness indeed increases automobile accidents in the morning and evening using a further approach.

6.5 Heterogeneity Analysis

Evaluating any possible heterogeneity of DST on total automobile accidents is also relevant. First, I analyze the effect of urban/rural locations on total accidents. Total urban accidents represent 85% of the total crashes and the main variability of automobile accidents. Second, I perform estimates for the country's northern, central, and southern parts since they present different sunrise and sunset times. For a uniform policy, I would expect to see some heterogeneity if different sunset and sunrise times occur at different moments. Moreover, I add an analysis by gender and by drivers

noisy. However, with this caveat in mind, the tables D.20 and D.21 show evidence in favour of the light mechanism when I analyze the remainder of the spring DST and fall ST and in general, the expected signs in all time horizons.

[40]Their specification controls for day-of-year, hour, day-of-week, holiday, year, and municipality fixed effects; therefore, their identification variation comes from comparing onset and offset of darkness throughout the year and by latitude. Also, the transition and the changes in dates when DST begins and ends exogenously reallocate light, allowing us to have an additional source of variation. In this case, darkness is a continuous variable from 0 to 1 that gauges the percent of sunlight in an hour.

[41]I also carry out this exercise of the impact of darkness in dates where my variation in dates sometimes cover by DST and sometimes not, where my identification variation to the longer-term comes (March 9th to May 15th and August 12th to October 15th). The results are similar to the main ones with all days. Finally, I also analyze the effect of darkness with the switching dates from above, and the results are the same.

behaving in risky ways, such as drinking alcohol and taking recreational drugs while driving[42].

In table D.3, Panel A and B show that the results align with the overall short-term null results. In the longer-term in Panel C, columns 1 and 4 show that the negative effect of DST is concentrated in urban and central zones of the country. These results make sense since urban accidents represent 85% of the total crashes, and also 73% of the accidents occur in the central geographic part of the country. The latter may explain the null results for north and south displayed in columns 3 and 5 of Panel C. Moreover, columns 1 to 3 in table D.5 also show null results by gender and crushes involving driver risky behaviours in the short-term. In panel C, I observe bigger longer-term effects for men but not for women drivers over the DST and an impact for spring DST and fall ST for the people driving under the influence of alcohol or recreational drugs. One reason that may explain why the longer-term results are only significant for men and at the same time, similar to the total accidents, could it be that men cause 80% of the motor vehicle crashes in the sample. In this line, Rhodes and Pivik 2011 present evidence that men behave more riskily than women while driving, and it may be the case that the positive (more light in the evening) or negative (less light in the evening) shocks exacerbate these patterns on men rather than women. Also, the estimates for women are noisier than men, and this prevents the finding of significant coefficients.

Additionally, table D.4 presents a test for the sleep disruption effect for the same definition of the groups from table D.3. Panels A and B show the results for the RDD and Panel C for the DID approach. These results show no heterogeneity by urban/rural or geographic area. Following the transition into DST, there is an increase in automobile accidents in the first week and a reduction of them after the transition into ST, which is in line with the main results of this mechanism.

Moreover, in columns 4 to 6 in table D.5, I observe evidence in favour of the sleep disruption channel for men, women, and for behaving riskily while driving. However, the RDD and DID show that the sleep disruption impact is higher for women than men at both transitions. The latter may be related to the idea that women need more sleep than men (Burgard and Ailshire 2013); therefore, the sleep patterns' shock might affects them more. I also analyze the effect of the transitions in driving under the influence of alcohol or recreational drugs, finding that the sleep pattern effect is heavily present, being symmetrical at both transitions.

Finally, in the appendix from figures E.16 to E.18, I present falsification tests based on permutation tests to show the validity of the significant estimates from the RDD method for the heterogeneity analysis. In general, almost all coefficients are significant at conventional levels.

7 Cost-Benefit Analysis and Robustness Checks

A relevant exercise in evaluating the DST policy is to provide evidence on the effects of these transitions on materialized costs, i.e. healthcare utilization, insurance costs, absence from work, etc, caused by automobile accidents. Doing an evaluation of this nature is challenging since indirect

[42] A complete set of tables for the RDD and DID approaches by each category of the heterogeneity analysis are available in the appendix from the tables D.24 to D.39.

costs are difficult to measure, such as the loss of productivity, mental health consequences, earning losses and the quality of health services, among others. However, I can do a back-of-the-envelope calculation based on the direct costs of each type of accident (treatment, administrative, and automobile damage) and the human cost loss. Nonetheless, this exercise omits the indirect costs described above. Moreover, it is calculated for only 131 days of the switching dates, almost 60% of the DST regime more or less. For these reasons, it must be considered a lower bound of the true avoided cost of implementing the DST regime on automobile accidents.

To do these calculations, I follow two methodologies. First, I calculate the average number of victims by the severity level (no injury, minor, serious, fatal) involved in each definition of accident[43] (minor, serious, fatal) and combining with information about the cost by victims/severity that is taken from CONASET[44], I calculate the average cost of each type/definition of accident (minor, serious and fatal). Then, I multiply the cost by the daily average number of accidents, by the significant coefficients and by the number of days that the coefficient represents by each type of accident[45]. I obtain that implementing the DST regime will reduce the cost of minor accidents to USD 331,153, the cost of serious accidents to USD 705,582, and the cost of fatal accidents to USD 162,068,943. The total cost reduction will be USD 163,105,679. Even if I consider that minor accidents may not be well measured due to under-reporting, the total cost reduction of serious and fatal accidents will be similar to the total calculated above since the cost of minor accidents is pretty small compared to the cost of serious or fatal accidents.

The second approach to calculate the reduction in cost of implementing the DST regime comes from Laliotis et al. 2023. Instead of estimating the effect of DST on automobile accidents, I estimate the impact of DST on the number of victims by the level of severity. Using the DID approach, I present these estimates in the appendix in the table D.40. I multiply the victim cost by the average daily number of victims[46], by the significant coefficient, and by the number of days that the coefficient represents for each type of victim/analysis. From these calculations, I determine that implementing the DST regime will reduce the cost of minor accidents to USD 268,730, serious accidents to USD 554,066 and fatal accidents to USD 108,526,434. The total cost reduction will be USD 109,349,230. Again, when I omit the minor accidents, the amount is similar to the total. The differences seem big if I compare the latter number with the one for the first approach. However, the disparity resides mainly since small changes in the count of fatal victims produce changes in the calculated amounts.

Finally, I carry out multiple robustness checks to my baseline results using different levels of

[43]For instance, an accident is considered fatal if one of the victims passes away. However, the same accident can have serious or minor victims as well. Serious accidents can have minor victims, and minor accidents only can have minor victims or can not have. The average number of victims/severity by each definition of accidents are available in the table C.2.

[44]Details of the cost by type of accidents are taken from https://www.conaset.cl/wp-content/uploads/2023/08/Costos-Siniestros-2022.pdf and SECTRA 2014. The costs are available in the table C.1

[45]I use the coefficients available in the appendix in the tables D.17, D.23 and D.21. The remainder of DST and ST represents the average effect over 50 and 53 days, respectively.

[46]In the table C.3, I present the average number of victims by level of severity of the DST or ST regime.

analysis, adjustments, and specifications. This analysis is presented in the appendix B alongside with an analysis of all types of automobile accidents. In general, the baseline results are robust to these checks.

8 Conclusions

Daylight Saving Time is a policy practiced by 70 countries around the world. In recent years, literature has blossomed, studying other outcomes apart from energy use, such as automobile accidents, well-being, productivity, and accidents at work. The DST-related literature shows that the effect of this policy in saving energy is very little or non-existent. For this reason, it is important to evaluate the policy holistically. I analyze the impact of the DST regime on the number of automobile accidents. Road accidents are an important subject to explore, given the high number of people involved in driving incidents daily and their direct relationship to labor productivity[47]. This research becomes more important in a developing country because, among other factors, access to health services is more precarious, amplifying any negative shocks of automobile accidents.

I follow the strategy proposed by Sexton and Beatty 2014, Doleac and Sanders 2015 and implemented for fatal automobile accidents by Smith 2016. Using the discrete nature of the transitions into DST and ST and the largest variation in dates covered by the policy in any paper so far, I find a null effect around the transition and a net decrease in automobile accidents over the dates covered by the DST regime in the longer-run. I analyze different portions of the day to disentangle the null effect in the short-run. I find that the sleep disruption effect translates into an increase (decrease) in the number of accidents in the first week between 3.9 to 6.1%(2.8 to 4%) just after the transition into DST(ST). I do not find conclusive evidence for the light mechanism using total accidents. After the sleep disruption effect dissipates, I find the expected sign in the evening but not in the morning. I provide suggestive evidence that this may be explain due to the under-reporting of minor accidents in the morning over the DST regime. I show that when I focus on serious+fatal accidents, the negative effects in the morning disappear, and the expected signs show up, supporting the light mechanism; while I analyze minor accidents, the negative signs in the morning keep turning up. Therefore, I show evidence that ambient light reduces serious and fatal accident risk. Finally, the overall reduction in total automobile accidents is mostly explained by accidents in urban locations, in the central part of the country, and caused by men driving. Additionally, the sleep disruption channel exists in almost all groups studied in the heterogeneity analysis, being consistently larger for women than men.

Regarding DST policy, these results should be taken as evidence to evaluate the efficiency of using two different time regimes during the year. I calculate a lower bound of the saving costs of implementing the DST over the dates where I identify the effect. It would save between 108 and 162 million USD annually. It is important to highlight that these transitions and their mechanisms affect other outcomes. For example, the light mechanism reduces crime (Doleac and Sanders 2015,

[47] On average, 42% of the cost of an automobile accident corresponds to a loss of productivity (Mello et al. 2009), and people involved in an accident lost between 0.3 and 2,202.4 days of work (Ebel et al. 2004).

Tealde 2021, Domınguez and Asahi 2023). Additionally, in the short-term, the transition into DST, through the alteration of the circadian rhythm, decreases productivity (Kamstra et al. 2000), increases heart attacks and mortality (Janszky et al. 2012, Toro et al. 2015, Cook 2022), reduces life satisfaction(Kountouris and Remoundou 2014, Costa et al. 2022) and increases the number of accidents at work (Barnes and Wagner 2009, Depalo 2023). However, the research is inconclusive since it has shown different results depending on the study context. Given this heterogeneity across countries, additional research in other contexts, such as developing countries, and other unexplored areas, such as school performance, is needed to holistically evaluate the policy.

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A Appendix: Explanation of Table 1

Following entering the DST regime, in the short-run, altering the circadian rhythm generates sleep of worse quality and of lesser duration. Therefore, during hours not affected by the reallocation of light, an increase in automobile accidents should be seen. In the morning, since there is less daylight and people are affected by sleep deprivation, an increase in accidents should be experienced. In the evening, the net effect is unclear since people face a safer environment and sleep deprivation. In the long-run, only the light effect is present. Therefore, automobile accidents should be reduced in the evening and increased in the morning. The net effect depends on the difference in risk levels between the morning and evening.

Following the ST transition, in the short-run, people experience poorer quality sleep, however, they may sleep more time. As a consequence, the net sleep effect is not clear. The light reallocation generates a safer environment in the morning and a riskier one in the evening. Hence, the sign of the effect is not clear in the morning, evening, and rest of the day. If the sleep disruption impact is positive, more accidents are expected in the evening, undetermined change in the morning, and more accidents the rest of the day. If the effect is negative, one would expect fewer in the morning, undetermined change in the evening, and fewer the rest of the day. In the long run, automobile accidents should decrease in the morning and increase in the evening. Again, the net effect depends on the difference in riskiness between the morning and evening hours.

B Robustness Checks and Analysis by Type of Automobile Accident

As robustness checks, first, I present the RDD and DID FE model adding municipality fixed effects. These fixed effects control for characteristics specific to municipalities that do not change over time and might affect automobile accidents, such as road and lightning conditions, availability of roads, car flow, weather conditions, etc. Since the unit of observations is accidents in a municipality in a day, my sample is populated by zeros. For this reason, I estimate the DID FE model using a Poisson model. The results are shown in tables D.8 and D.9, which are similar to the main results. Second, in tables D.6 and D.7, I display the DST/ST transition effect on automobile accidents by adding the hour dimension. Estimates from the RDD and DID FE methods remain similar to the main results[48].

Additionally, I present the RDD estimates for different kernels (uniforms and epanechnikov) and different polynomial orders of the function $g()$. Also, I perform some exercises not adjusting the day when one hour is gained or lost and two additional corrections for the gained/missing hours, first multi-playing the day after the spring transition by 24/23 and 12/11 and the day before the fall transition by 24/25 and 12/13 for all-day hours and least light impacted hours respectively (following Janszky et al. 2012), and second, dropping the transition date (donut RDD). The null results found in the main estimates for all day are similar to the ones in tables D.10 and D.11. In terms of the last light-impacted hours, tables D.12 and D.13 provide evidence of the robustness of the sleep disruption effect, which is similar to the main results after the spring and fall transitions. Moreover, table D.14 shows the lack and difference in adjusting for the day before and following the fall and spring transition. The coefficient associated with the first week of DST and the first week of ST in columns 4 to 6 are similar in magnitude and sign to the main ones.

[48] I also analyze the DID FE model with Day-of-Year/Hour fixed effect, and results remain similar. This table is available upon request.

Finally, from tables D.16 to D.19 I show the RDD and DID estimates for each type of accident: minor, serious and fatal (and serious+fatal). This analysis shows me that when I focus on accidents that are less likely to be under-reported (serious or fatal), the results align better with the hypothesized mechanisms. Tables D.16 and D.17 use minor accidents as dependent variable and replicate the overall results since minor accidents represent 88% of all automobile accidents. I find that minor accidents are reduced by 2.2% in the DST regime. I also find evidence for the sleep mechanism and partial evidence for the light mechanism (evening results support it but morning results do not). In the case of serious car accidents, tables D.22 and D.23 show evidence in favour of both channels. Overall, I find that the DST regime reduces serious accidents in a 7.7%. Finally, D.20 and D.21 provide stronger evidence for the light mechanism. Less clear is the sleep disruption effect, which can be supported in column 5 in table D.21. Nevertheless, given the noisy estimates in both panels in columns 7 and 8 in table D.20, it is more difficult to find large estimates to be significant. The overall effect of the DST regime on fatal automobile accidents is a reduction of 7.5% over the remainder of the DST and a 11% reduction over the fall DST (or a increase of 11% over the fall ST).

C Cost-Benefit Analysis

In this section, I provide details of the cost-benefit analysis. The following table shows the average cost by victims and the type of injury used in both methods.

Table C.1: Average Cost of Victim by Severity

	Costs (Victim/USD)			
	Treatment	Administrative	Human	Car Damage
Minor	155.4	951.6		533.6
Serious	3,377	1,928.3		533.6
Fatal	628.6	3,437.8	3,247,786	533.6

Source: CONASET and SECTRA 2014. I use UF and dollar values at 07-23-2024 that are 37,588 and 946 Chilean pesos respectively to transform the costs to USD.

In the first method, I calculate the average number of victims by injury level and type of accident in the 64 first days of DST in spring and the 67 first days of ST in fall. In other words, an accident is considered fatal when one person involved passes away. However, more people involved in the same accident can end up having serious, minor, or no injuries. In the case of serious accidents, people can also have minor or no injuries. Finally, in minor accidents, people involved can have at least minor or no injuries.

Table C.2: Average Number of Victims by Severity and Type of Accident

	Spring DST			Fall ST		
Accident/N Victims	Minor	Serious	Fatal	Minor	Serious	Fatal
Minor	0.73			0.72		
Serious	0.74	1.11		0.67	1.12	
Fatal	0.73	0.32	1.11	0.71	0.32	1.10

Source: Own calculations based on CONASET data.

The last piece to calculate the cost-benefit analysis is the average number of daily accidents for the first approach and the average number of victims by level of severity for the second approach. These averages are calculated in the first 64 days of DST and the first 67 days of ST since they are the common support for estimating the longer-term effects of DST and ST. They are shown in the next table.

Table C.3: Average Number of Victims and Accidents by Severity

	Spring DST	Fall ST
Panel A: Accidents		
Minor	145.8	155.5
Serious	15.6	16.9
Fatal	3.6	4.2
Panel B: Victims		
Minor	120.7	126.0
Serious	18.7	20.2
Fatal	4.2	4.7

Source: Own calculations based on CONASET data.

D Appendix: Tables

Table D.1: The Effect of Darkness on Automobile Accidents

	(1) All Hours	(2) 0am-6am	(3) 7am-12pm	(4) 1pm-6pm	(5) 7pm-11pm
Panel A: All					
Darkness	0.114***	0.107**	0.255***	0.111***	0.057***
	(0.0128)	(0.0485)	(0.0219)	(0.0177)	(0.0109)
Panel B: Minor					
Darkness	0.105***	0.120**	0.254***	0.102***	0.043***
	(0.0128)	(0.0517)	(0.0222)	(0.0181)	(0.0108)
Panel C: Serious					
Darkness	0.233***	0.292**	0.251***	0.287***	0.189***
	(0.0281)	(0.143)	(0.0578)	(0.0447)	(0.0381)
Panel D: Fatal					
Darkness	0.666***	-0.242	0.204*	0.553***	1.057***
	(0.0621)	(0.203)	(0.118)	(0.120)	(0.0873)

Note: The dependent variable is the daily count of automobile accidents. Darkness is a continuous variables from zero to 1 who gauges the percentage of darkness in a single hour of the day by municipality. The regression also controls for day-of-week, holiday, day-of-year, year, hour of the day and municipality FEs. Estimates include all days between 01/01/2002 and 12/31/2018. I use a Poisson count model (PPML) to estimate the coefficients. Collumn 1 is a single regression and from columns 2 to 5, coefficients are estimated in one regression interacting the time frames with darkness. Standard errors are in parentheses and are clustered at the municipality level. *** p<0.01, ** p<0.05, * p<0.1

Table D.2: Difference-in-Difference estimates by different portions of the day

	Morning			Evening			Least light impacted		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
First week of DST	-0.0665**	-0.0692**	-0.0665**	-0.0386**	-0.0398**	-0.0388**	0.0360*	0.0379**	0.0394**
	(0.0278)	(0.0280)	(0.0276)	(0.0193)	(0.0196)	(0.0191)	(0.0187)	(0.0188)	(0.0186)
Second week of DST	-0.0606**	-0.0627**	-0.0607**	-0.0323*	-0.0333*	-0.0325*	0.0201	0.0230	0.0244
	(0.0258)	(0.0259)	(0.0256)	(0.0177)	(0.0178)	(0.0174)	(0.0182)	(0.0180)	(0.0178)
Remainder Spring DST	-0.0617***	-0.0650***	-0.0619***	-0.0565***	-0.0582***	-0.0569***	0.0017	0.0051	0.0074
	(0.0180)	(0.0183)	(0.0172)	(0.0135)	(0.0139)	(0.0128)	(0.0124)	(0.0124)	(0.0117)
First week of ST	0.0256	0.0275	0.0269	0.0018	0.0040	0.0025	-0.0301	-0.0244	-0.0284
	(0.0241)	(0.0242)	(0.0238)	(0.0182)	(0.0183)	(0.0180)	(0.0185)	(0.0187)	(0.0183)
Second week of ST	0.0585***	0.0607***	0.0600***	0.0219	0.0242	0.0228	0.0020	0.0081	0.0040
	(0.0227)	(0.0226)	(0.0223)	(0.0178)	(0.0178)	(0.0175)	(0.0175)	(0.0177)	(0.0174)
Remainder Fall ST	-0.0007	0.0021	0.0010	0.0416***	0.0446***	0.0426***	0.0037	0.0110	0.0057
	(0.0180)	(0.0184)	(0.0174)	(0.0130)	(0.0130)	(0.0125)	(0.0133)	(0.0136)	(0.0128)
1 week before DST		-0.0127			-0.0042			-0.0055	
		(0.0253)			(0.0170)			(0.0197)	
1 week before ST		0.0027			0.0082			0.0232	
		(0.0248)			(0.0181)			(0.0170)	
2 weeks before DST	-0.0007			0.0011			-0.0213		
	(0.0249)			(0.0186)			(0.0171)		
2 weeks before ST	-0.0092			-0.0052			-0.0157		
	(0.0245)			(0.0171)			(0.0179)		
Observations	6,209	6,209	6,209	6,209	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.720	0.720	0.721	0.762	0.762	0.762	0.780	0.780	0.780

Note: The dependent variable is the log of automobile crashes; all specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of Spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in each municipality. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.3: Heterogeneity Analysis: All Hours

	Urban (1)	Rural (2)	North (3)	Center (4)	South (5)
Panel A: RD Spring DST					
Transition	-0.0010	0.0211	0.0109	-0.0093	0.0288
	(0.0132)	(0.0316)	(0.0327)	(0.0156)	(0.0256)
Panel B: RD Fall ST					
Transition	-0.0035	0.0195	-0.0310	-0.0045	0.0288
	(0.0158)	(0.0261)	(0.0302)	(0.0175)	(0.0218)
Panel C: DID					
DST	-0.0223**	-0.0060	-0.0223	-0.0259**	-0.0221
	(0.00930)	(0.0181)	(0.0194)	(0.0103)	(0.0161)
ST	0.0081	0.0248	-0.0057	0.0144	0.0094
	(0.00902)	(0.0175)	(0.0189)	(0.00998)	(0.0156)
Observations	6,209	6,209	6,209	6,209	6,209

Note: The dependent variable is the daily log of automobile accidents. Panel A and B control for the daily price of gasoline, day-of-week, holiday, month, and year FE. Panel C controls for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before January 1st. fall DST is an indicator variable equal to one if the day falls under DST and occurs after January 1st. Least light impacted hours are those hours that are not not contained around +/- three hours from the average sunset and sunrise in every location. Panel A and B use the common mean squared error (MSE)-optimal bandwidth selector developed by Calonico et al. 2017. North correspond to I, II, III, IV and XV regions. Center corresponds to V, VI, VII and Metropolitan regions. South corresponds to VIII, IX, X, XI, XII and XIV regions. Estimates include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.4: Heterogeneity Analysis: Least Light Impacted Hours

	Urban	Rural	North	Center	South
	(1)	(2)	(3)	(4)	(5)
Panel A: RD Spring DST					
Transition	0.0561***	0.104**	0.0190	0.0642***	0.0596*
	(0.0199)	(0.0435)	(0.0394)	(0.0217)	(0.0357)
Panel B: RD Fall ST					
Transition	-0.0263	-0.102***	-0.0847**	-0.0359	-0.0179
	(0.0213)	(0.0343)	(0.0372)	(0.0241)	(0.0307)
Panel C: DID					
First week of DST	0.0384*	0.0383	-0.0062	0.0415*	0.0491
	(0.0201)	(0.0387)	(0.0409)	(0.0229)	(0.0320)
Second week of DST	0.0266	-0.0028	-0.0504	0.0380	0.0244
	(0.0205)	(0.0393)	(0.0401)	(0.0233)	(0.0348)
Remainder of Spring DST	0.0109	-0.0161	-0.0206	0.0079	0.0101
	(0.0133)	(0.0268)	(0.0259)	(0.0152)	(0.0223)
First week of ST	-0.0262	-0.0470	-0.0274	-0.0279	-0.0439
	(0.0192)	(0.0428)	(0.0392)	(0.0218)	(0.0331)
Second week of ST	0.0168	-0.0692*	-0.00208	-0.0057	0.0058
	(0.0195)	(0.0407)	(0.0393)	(0.0222)	(0.0350)
Remainder of Fall ST	0.0106	-0.0262	-0.0175	0.00627	0.0115
	(0.0133)	(0.0273)	(0.0280)	(0.0151)	(0.0229)
Observations	6,209	6,209	6,209	6,209	6,209

Note: The dependent variable is the daily log of automobile accidents or the daily count. Panel A and B control for the daily price of gasoline, day-of-week, holiday, month, and year FE. Panel C controls for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before January 1st. fall DST is an indicator variable equal to one if the day falls under DST and occurs after January 1st. Least light impacted hours are those hours that are not not contained around +/- three hours from the average sunset and sunrise in every location. Panel A and B use the common mean squared error (MSE)-optimal bandwidth selector developed by Calonico et al. 2017. North correspond to I, II, III, IV and XV regions. Center corresponds to V, VI, VII and Metropolitan regions. South corresponds to VIII, IX, X, XI, XII and XIV regions. Panel C, columns 2, 3 and 5 use a Poisson count model (PPML) to estimate the coefficients. Panel A and B, in columns 2, 3 and 5 use as dependent variable the daily log of automobile accidents plus one. Estimates include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.5: Heterogeneity Analysis: Gender and Drunk Driving

	All Hours			Least Light Impacted Hours		
	Men	Women	Drunk Driving	Men	Women	Drunk Driving
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: RD Spring DST						
Transition	-0.0111	0.0205	0.0575	0.0546***	0.0740**	0.0948*
	(0.0129)	(0.0189)	(0.0452)	(0.0208)	(0.0333)	(0.0533)
Panel B: RD Fall DST						
Transition	0.00435	0.0114	0.0040	-0.0385*	-0.0422	-0.0812*
	(0.0138)	(0.0246)	(0.0390)	(0.0209)	(0.0341)	(0.0484)
Panel C: DID						
First week of DST	-0.0158	-0.00694	0.0099	0.0341*	0.0777**	0.0456
	(0.0150)	(0.0267)	(0.0396)	(0.0205)	(0.0329)	(0.0432)
Second week of DST	-0.0235	0.0058	-0.0143	0.0270	0.0162	-0.0159
	(0.0153)	(0.0272)	(0.0410)	(0.0209)	(0.0361)	(0.0517)
Remainder of Spring DST	-0.0326***	-0.0242	-0.0768**	0.0041	0.0207	-0.0467
	(0.00994)	(0.0177)	(0.0337)	(0.0136)	(0.0223)	(0.0364)
First week of ST	-0.0027	-0.0105	0.0373	-0.0256	-0.0279	-0.0614
	(0.0143)	(0.0254)	(0.0443)	(0.0195)	(0.0331)	(0.0541)
Second week of ST	0.0289**	0.0001	0.0285	0.0075	-0.0125	-0.0373
	(0.0145)	(0.0258)	(0.0497)	(0.0198)	(0.0341)	(0.0580)
Remainder of Fall ST	0.0218**	-0.0150	0.126***	0.0117	-0.0102	0.108***
	(0.00992)	(0.0176)	(0.0332)	(0.0135)	(0.0223)	(0.0369)
Observations	6,209	6,209	6,209	6,209	6,209	6,209

Note: The dependent variable is the daily log of automobile accidents or the daily count. Panel A and B control for the daily price of gasoline, day-of-week, holiday, month, and year FE. Panel C controls for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before January 1st. fall DST is an indicator variable equal to one if the day falls under DST and occurs after January 1st. Least light impacted hours are those hours that are not not contained around +/- three hours from the average sunset and sunrise in every location. Panel A and B use the common mean squared error (MSE)-optimal bandwidth selector developed by Calonico et al. 2017. Panel C, columns 3, 5 and 6 use a Poisson count model (PPML) to estimate the coefficients. Panel A and B, in columns 3, 5 and 6 use as dependent variable the daily log of automobile accidents plus one. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.6: RDD by Hour of the Day

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.00519	-0.00795	-0.0851***	-0.100***	-0.0224	-0.0194	0.0612***	0.0629***
	(0.0120)	(0.0112)	(0.0279)	(0.0308)	(0.0175)	(0.0179)	(0.0210)	(0.0206)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	63	68	54	40	73	72	42	47
Bandwidth Right	63	74	54	48	73	67	42	42
Observations	45636	51418	11020	9010	14091	13421	13777	14575
Panel B: Fall ST								
Transition	0.00641	0.00859	0.0619**	0.0583**	0.0104	0.0141	-0.0297*	-0.0277
	(0.0140)	(0.0146)	(0.0259)	(0.0276)	(0.0222)	(0.0225)	(0.0180)	(0.0182)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	48	39	67	59	46	42	53	45
Bandwidth Right	48	53	67	60	46	49	53	67
Observations	33684	32124	13300	11810	8587	8516	16797	17865

Note: The dependent variable is the log of the daily number of automobile accidents. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, hour, month, and year FE. Morning hours are from 4am to 10am. Evening hours are from 3pm to 8pm. Least Light Impacted Hours are the remaining hours of the day. Panel A and B use the common mean squared error (MSE)-optimal bandwidth selector developed by Calonico et al. 2017 Estimates include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.7: DID FE Model by Hour of the Day

	All	Morning	Evening	Least Light Impacted
	(1)	(2)	(3)	(4)
First week of DST	-0.0225	-0.104***	-0.0478**	0.0434**
	(0.0138)	(0.0299)	(0.0215)	(0.0199)
Second week of DST	-0.0189	-0.0547*	-0.0409*	0.0180
	(0.0141)	(0.0306)	(0.0219)	(0.0202)
Remainder of Spring DST	-0.0454***	-0.0838***	-0.0652***	-0.0105
	(0.00872)	(0.0189)	(0.0135)	(0.0126)
First week of ST	0.00244	0.0380	0.00374	-0.0212
	(0.0133)	(0.0286)	(0.0205)	(0.0192)
Second week of ST	0.0275**	0.0407	0.0445**	0.00893
	(0.0134)	(0.0289)	(0.0208)	(0.0193)
Remainder of Spring ST	0.0241***	0.0231	0.0526***	0.00839
	(0.00871)	(0.0188)	(0.0135)	(0.0126)
	140,257	39,651	37,206	63,400
Observations	0.524	0.432	0.356	0.544

Note: The dependent variable is the log of the daily number of automobile accidents ot the daily count. All estimates control for the daily price of gasoline, day-of-year, hour, day-of-week, holiday, and year FE. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before January 1st. fall DST is an indicator variable equal to one if the day falls under DST and occurs after January 1st. Morning hours are from 4am to 10am. Evening hours are from 3pm to 8pm. Least Light Impacted Hours are the remaining hours of the day. Estimates include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.8: RDD with Municipality Fixed Effects

	All	Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)
Panel A: Spring DST				
Transition	-0.0000	-0.0051***	-0.0011	0.0042***
	(0.0019)	(0.0013)	(0.0013)	(0.0015)
Bandwidth	50	45	58	48
Observations	2,000,360	2,000,360	2,000,360	2,000,360
Panel B: Fall ST				
Transition	-0.0008	0.0016	0.0008	-0.0034*
	(0.0024)	(0.0011)	(0.0017)	(0.0018)
Bandwidth	34	66	37	35
Observations	1,913,672	2,007,584	2,007,584	2,007,584

Note: The dependent variable is the log of the daily number of automobile accidents plus one. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, month, year dummies, and municipality fixed effects. Panel A and B use the common mean squared error (MSE)-optimal bandwidth selector developed by Calonico et al. 2017 Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses and at the municipality level. *** p<0.01, ** p<0.05, * p<0.1

Table D.9: DID FE Model with Municipality Fixed Effects

	All	Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)
First week of DST	-0.0196*	-0.0570***	-0.0412**	0.0344**
	(0.0107)	(0.0183)	(0.0173)	(0.0162)
Second week of DST	-0.0237**	-0.0522***	-0.0404**	0.0128
	(0.0104)	(0.0195)	(0.0196)	(0.0149)
Remainder of Spring DST	-0.0388***	-0.0449***	-0.0608***	-0.0045
	(0.00856)	(0.0144)	(0.0115)	(0.0122)
First week of ST	-0.0001	0.0259	0.0123	-0.0258
	(0.0104)	(0.0171)	(0.0134)	(0.0174)
Second week of ST	0.0212*	0.0456**	0.0229	0.0022
	(0.0112)	(0.0192)	(0.0172)	(0.0155)
Remainder of Fall ST	0.0167*	-0.0111	0.0458***	0.0070
	(0.00944)	(0.0133)	(0.0135)	(0.0127)
Observations	2,125,232	2,125,232	2,125,232	2,125,232
$Adj - R^2$	0.282	0.259	0.259	0.267

Note: The dependent variable is the daily number of automobile accidents. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, year dummies, and municipality fixed effects. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before January 1st. fall DST is an indicator variable equal to one if the day falls under DST and occurs after January 1st. Least light impacted hours are those hours that are not not contained around +/- three hours from the average sunset and sunrise in every location. All estimates use a Poisson count model (PPML). Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses and at the municipality level. *** p<0.01, ** p<0.05, * p<0.1

Table D.10: RDD: All Hours

	Alternative Kernels			No adjust	Adj. missing hour	No transition
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Spring DST						
Transition	-0.0059	0.0077	0.0061	-0.0082	-0.0055	-0.0104
	(0.0120)	(0.0144)	(0.0140)	(0.0121)	(0.0120)	(0.0124)
Kernel	Tri	Uni	Epa	Tri	Tri	Tri
Bandwidth	81	49	56	80	81	78
Observations	2610	1584	1808	2577	2610	2496
Panel B: Fall ST						
Transition	0.006	0.0058	0.0000	0.0027	0.0074	0.0036
	(0.0130)	(0.0164)	(0.0156)	(0.0131)	(0.0129)	(0.0134)
Kernel	Tri	Uni	Epa	Tri	Tri	Tri
Bandwidth	65	35	42	63	66	64
Observations	2031	1101	1318	1969	2062	1985

Note: The dependent variable is the daily log of automobile accidents. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. Tri means Triangular kernel, Epa means Epanechnikov Kernel, Uni means uniform kernel. No adjust means the the 1 am hour accident of the first day of DST is not adjusted by the missing hour or the additional hour at 11pm the day before the transition into ST. Adj. Missing hour means that the transition day into DST is adjusted by 24/23 and the day before the transition into ST is adjusted by 24/25. No transition means the the transition date is dropped. Optimal bandwidth is chosen using Calonico, Cattaneo and Titiunik method. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.11: RDD: Alternatives Polynomials All Hours

	(1)	(2)	(3)	(4)	(5)	(6)
	Spring DST			Fall ST		
Transition	-0.0255	-0.0886***	-0.107***	-0.0056	-0.0038	0.0001
	(0.0206)	(0.0273)	(0.0306)	(0.0172)	(0.0211)	(0.0252)
Polynomial Order	2	3	4	2	3	4
Bandwidth	55	54	67	79	91	100
Observations	1776	1744	2160	2454	2812	3059

Note: The dependent variable is the daily log of automobile accidents. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Bandwidth selectors is a common mean squared error (MSE)-optimal bandwidth selector that uses a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.12: RDD: Least Light Impacted Hours

	Alternative Kernels			No adjust	Adj. missing hour	No transition
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Spring DST						
Transition	0.0601***	0.0755***	0.0638***	0.0504***	0.0518***	0.0565***
	(0.0194)	(0.0214)	(0.0200)	(0.0194)	(0.0190)	(0.0208)
Kernel	Tri	Uni	Epa	Tri	Tri	Tri
Bandwidth	45	31	40	46	47	47
Observations	1456	1008	1296	1472	1520	1520
Panel B: Fall ST						
Transition	-0.0397*	-0.0302	-0.0283*	-0.0399*	-0.0471**	-0.0434**
	(0.0204)	(0.0215)	(0.0164)	(0.0215)	(0.0207)	(0.0191)
Kernel	Tri	Uni	Epa	Tri	Tri	Tri
Bandwidth	44	33	66	43	43	43
Observations	1424	1072	2128	1376	1392	1392

Note: The dependent variable is the daily log of automobile accidents in the least light impacted hours. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. Tri means Triangular kernel, Epa means Epanechnikov Kernel, Uni means uniform kernel. No adjust means the the 1 am hour accident of the first day of DST is not adjusted by the missing hour or the additional hour at 11pm the day before the transition into ST. Adj. Missing hour means that the transition day into DST is adjusted by 12/11 and the day before the transition into ST is adjusted by 12/13. No transition means the the transition date is dropped. Optimal bandwidth is chosen using Calonico, Cattaneo and Titiunik method. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.13: RDD: Alternatives Polynomials Least Light Impacted Hours

	(1)	(2)	(3)	(4)	(5)	(6)
	Spring DST			Fall ST		
Transition	0.0531**	0.0485	0.0693*	-0.0493**	-0.0543**	-0.0416
	(0.0239)	(0.0341)	(0.0371)	(0.0227)	(0.0261)	(0.0322)
Polynomial Order	2	3	4	2	3	4
Bandwidth	62	52	68	76	100	100
Observations	2000	1680	2192	2440	3159	3159

Note: The dependent variable is the daily log of automobile accidents in the least light impacted hours. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, and year FE. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Bandwidth selectors is a common mean squared error (MSE)-optimal bandwidth selector that uses a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.14: Difference-in-Difference FE: Different Adjustments

	All Hours			Least Light Impacted Hours		
	No Adjust	Adj Missing	No Transition	No Adjust	Adj Missing	No Transition
	(1)	(2)	(3)	(4)	(5)	(6)
First week of DST	-0.0205	-0.0141	-0.0219	0.0404**	0.0273	0.0279
	(0.0144)	(0.0144)	(0.0154)	(0.0189)	(0.0189)	(0.0202)
Second week of DST	-0.0190	-0.0186	-0.0186	0.0244	0.0236	0.0236
	(0.0146)	(0.0146)	(0.0146)	(0.0192)	(0.0192)	(0.0193)
Remainder of Spring DST	-0.0331***	-0.0332***	-0.0323***	0.0072	0.0075	0.0080
	(0.0095)	(0.0095)	(0.0095)	(0.0125)	(0.0125)	(0.0125)
First week of ST	-0.0047	-0.0034	-0.0036	-0.0277	-0.0306*	-0.0261
	(0.0137)	(0.0137)	(0.0137)	(0.0180)	(0.0180)	(0.0181)
Second week of ST	0.0230*	0.0238*	0.0242*	0.0044	0.0027	0.0057
	(0.0139)	(0.0139)	(0.0139)	(0.0183)	(0.0183)	(0.0183)
Remainder of Fall ST	0.0142	0.0155	0.0156	0.0062	0.0035	0.0077
	(0.0095)	(0.0095)	(0.0096)	(0.0125)	(0.0125)	(0.0126)
Observations	6,209	6,209	6,177	6,209	6,209	6,177
$Adj - R^2$	0.845	0.845	0.845	0.780	0.780	0.780

Note: The dependent variable is the daily number of automobile accidents. All estimates control for the daily price of gasoline, day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before January 1st. fall DST is an indicator variable equal to one if the day falls under DST and occurs after January 1st. No adjust means the the 1 am hour accident of the first day of DST is not adjusted by the missing hour or the additional hour at 11pm the day before the transition into ST. Adj. Missing means that the transition day into DST is adjusted by 12/11 and 24/23 for least light impacted hours respectively and all hours and the day before the transition into ST is adjusted by 12/13 and 24/25 for least light impacted hours and all hours respectively. No transition means the the transition date is dropped. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.15: RDD without Month FE

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0083	-0.0089	-0.113***	-0.135***	-0.0224	-0.0088	0.0502**	0.0592***
	(0.0123)	(0.0122)	(0.0304)	(0.0307)	(0.0151)	(0.0162)	(0.0211)	(0.0201)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	77	77	38	29	81	79	38	49
Bandwidth Right	77	80	38	52	81	61	38	36
Observations	2480	2529	1232	1312	2610	2256	1232	1376
Panel B: Fall ST								
Transition	0.0000	-0.0097	0.0491*	0.0528**	0.0056	0.0006	-0.0342*	-0.0324
	(0.0179)	(0.0173)	(0.0272)	(0.0265)	(0.0205)	(0.0227)	(0.0207)	(0.0203)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	34	31	40	45	47	33	42	39
Bandwidth Right	34	45	40	42	47	48	42	55
Observations	1070	1201	1296	1408	1520	1312	1360	1520

Note: The dependent variable is the log of automobile accidents. All estimates control for the daily price of gasoline, day-of-week, holiday, year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in automobile accidents that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in each municipality. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.16: RDD: Minor Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0038	0.0042	-0.0741***	-0.0886***	-0.0009	-0.0008	0.0517**	0.0456**
	(0.0124)	(0.0134)	(0.0280)	(0.0282)	(0.0167)	(0.0168)	(0.0203)	(0.0179)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	84	77	49	39	78	76	46	51
Bandwidth Right	84	64	49	67	78	78	46	77
Observations	2709	2272	1584	1712	2512	2480	1488	2064
Panel B: Fall ST								
Transition	-0.0026	-0.0049	0.0730***	0.0751***	-0.0107	-0.0201	-0.0360*	-0.0317
	(0.0148)	(0.0168)	(0.0216)	(0.0240)	(0.0192)	(0.0225)	(0.0200)	(0.0213)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	53	36	70	61	60	38	50	39
Bandwidth Right	53	48	70	51	60	52	50	53
Observations	1659	1324	2254	1808	1936	1456	1616	1488

Note: The dependent variable is the log of the daily number of automobile accidents. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.17: DID Model: Minor Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0216**				
	(0.00920)				
First week of DST		-0.0107	-0.0699**	-0.0184	0.0354*
		(0.0153)	(0.0280)	(0.0207)	(0.0203)
Second week of DST		-0.0141	-0.0765***	-0.00946	0.0282
		(0.0155)	(0.0285)	(0.0211)	(0.0206)
Remainder of Spring DST		-0.0272***	-0.0664***	-0.0403***	0.0127
		(0.0101)	(0.0185)	(0.0137)	(0.0134)
Fall ST	0.0090				
	(0.00892)				
First week of ST		-0.0022	0.0414	-0.0036	-0.0285
		(0.0145)	(0.0267)	(0.0197)	(0.0193)
Second week of ST		0.0162	0.0693**	0.0024	-0.0019
		(0.0148)	(0.0271)	(0.0200)	(0.0196)
Remainder of Fall ST		0.0122	0.0130	0.0203	0.0090
		(0.0101)	(0.0185)	(0.0137)	(0.0134)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.857	0.857	0.745	0.776	0.780

Note: The dependent variable is the daily log of automobile crashes. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.18: RDD: Serious+Fatal Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0166	-0.0094	-0.0362	-0.0346	-0.0989***	-0.107***	0.0841**	0.0984**
	(0.0248)	(0.0260)	(0.0425)	(0.0385)	(0.0362)	(0.0338)	(0.0390)	(0.0418)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	71	57	62	54	65	64	59	51
Bandwidth Right	71	68	62	102	65	86	59	50
Observations	2288	2016	2000	2535	2096	2423	1904	1632
Panel B: Fall ST								
Transition	0.0192	0.0133	-0.0602	-0.0547	0.116***	0.102**	-0.0444	-0.0666*
	(0.0223)	(0.0272)	(0.0395)	(0.0413)	(0.0415)	(0.0406)	(0.0376)	(0.0376)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	87	57	82	69	48	76	53	42
Bandwidth Right	87	59	82	81	48	38	53	73
Observations	2694	1815	2595	2415	1552	1832	1712	1856

Note: The dependent variable is the log of the daily number of automobile accidents in columns 1 and 2 and the log of the daily number of automobile accidents plus one for the remaining columns. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.19: DID Model: Serious+Fatal Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0747***				
	(0.0179)				
First week of DST		-0.0485	0.0093	-0.151***	0.0601
		(0.0297)	(0.0584)	(0.0419)	(0.0435)
Second week of DST		-0.0512*	0.0868*	-0.193***	-0.0009
		(0.0302)	(0.0526)	(0.0416)	(0.0488)
Remainder of Spring DST		-0.0891***	-0.0079	-0.160***	-0.0401
		(0.0197)	(0.0372)	(0.0285)	(0.0282)
Fall ST	0.0211				
	(0.0174)				
First week of ST		-0.0236	-0.0602	0.0133	-0.0374
		(0.0283)	(0.0529)	(0.0408)	(0.0383)
Second week of ST		0.0624**	-0.0181	0.142***	0.0461
		(0.0287)	(0.0505)	(0.0393)	(0.0420)
Remainder of Fall ST		0.0279	-0.0968***	0.162***	-0.0147
		(0.0196)	(0.0353)	(0.0277)	(0.0269)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.313	0.314	0.0413	0.0405	0.117

Note: The dependent variable the daily log of automobile serious and fatal crashes in columns 1 and 2 and the daily count for the remaining columns. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Columns 3, 4 and 5 use a Poisson count model (PPML) to estimate the coefficients. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.20: RDD: Fatal Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0463	-0.0490	-0.0625	-0.0469	-0.0924*	-0.0892*	0.0006	-0.0016
	(0.0403)	(0.0388)	(0.0547)	(0.0530)	(0.0501)	(0.0456)	(0.0397)	(0.0424)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	71	64	47	46	54	51	88	78
Bandwidth Right	71	97	47	55	54	98	88	74
Observations	2,288	2,610	1,520	1,632	1,744	2,419	2,841	2,448
Panel B: Fall ST								
Transition	0.0606	0.0617	-0.0692	-0.0705	0.230***	0.234***	-0.0164	-0.0449
	(0.0449)	(0.0432)	(0.0456)	(0.0444)	(0.0476)	(0.0493)	(0.0412)	(0.0477)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	64	73	64	71	63	53	84	61
Bandwidth Right	64	66	64	64	63	67	84	62
Observations	2,064	2,162	2,064	2,173	2,032	1,936	2,688	1,984

Note: The dependent variable is the log of the daily number of automobile accidents plus one. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.21: DID Model: Fatal Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0509				
	(0.0357)				
First week of DST		-0.0319	-0.0858	-0.252**	0.153*
		(0.0563)	(0.142)	(0.107)	(0.0828)
Second week of DST		0.0133	0.409***	-0.306***	0.0476
		(0.0593)	(0.114)	(0.103)	(0.0947)
Remainder of Spring DST		-0.0751*	0.161*	-0.373***	0.0195
		(0.0397)	(0.0841)	(0.0688)	(0.0574)
Fall ST	0.109***				
	(0.0327)				
First week of ST		0.0673	-0.153	0.291***	0.0128
		(0.0537)	(0.113)	(0.0885)	(0.0779)
Second week of ST		0.128**	-0.0960	0.405***	0.0260
		(0.0528)	(0.111)	(0.0890)	(0.0833)
Remainder of Fall ST		0.123***	-0.142*	0.342***	0.0726
		(0.0368)	(0.0791)	(0.0625)	(0.0551)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.0746	0.0747	0.0532	0.0503	0.0793

Note: The dependent variable is the daily number of fatal automobile crashes. All regressions use a Poisson count model (PPML) to estimate the coefficients. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.22: RDD: Serious Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0165	-0.0053	-0.0354	-0.0379	-0.0952***	-0.0917***	0.110**	0.112**
	(0.0270)	(0.0291)	(0.0421)	(0.0415)	(0.0339)	(0.0353)	(0.0447)	(0.0453)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	76	60	75	78	84	79	51	49
Bandwidth Right	76	69	75	77	84	76	51	50
Observations	2448	2080	2416	2496	2709	2496	1648	1600
Panel B: Fall ST								
Transition	-0.0006	0.0027	-0.0314	-0.0326	0.0574	0.0428	-0.0384	-0.0486
	(0.0309)	(0.0321)	(0.0413)	(0.0454)	(0.0491)	(0.0464)	(0.0407)	(0.0413)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	62	59	90	77	44	75	58	49
Bandwidth Right	62	56	90	71	44	35	58	66
Observations	1938	1797	2874	2375	1424	1769	1872	1856

Note: The dependent variable is the log of the daily number of serious automobile accidents in columns 1 and 2 , and the log of the daily number of automobile accidents plus one for the remaining columns. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.23: DID Model: Serious Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0765***				
	(0.0200)				
First week of DST		-0.0476	0.0289	-0.126***	0.0363
		(0.0331)	(0.0631)	(0.0486)	(0.0477)
Second week of DST		-0.0635*	0.0120	-0.166***	-0.0118
		(0.0337)	(0.0622)	(0.0454)	(0.0527)
Remainder of Spring DST		-0.0893***	-0.0457	-0.118***	-0.0481
		(0.0219)	(0.0409)	(0.0311)	(0.0315)
Fall ST	-0.0036				
	(0.0194)				
First week of ST		-0.0534*	-0.0382	-0.0449	-0.0528
		(0.0315)	(0.0619)	(0.0444)	(0.0428)
Second week of ST		0.0412	-0.0027	0.0858*	0.0520
		(0.0320)	(0.0560)	(0.0448)	(0.0475)
Remainder of Fall ST		0.0037	-0.0845**	0.122***	-0.0423
		(0.0219)	(0.0391)	(0.0306)	(0.0304)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.263	0.264	0.0452	0.0357	0.0943

Note: The dependent variable the daily log of automobile serious crashes in columns 1 and 2 and the daily count for the remaining columns. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Columns 3, 4 and 5 use a Poisson count model (PPML) to estimate the coefficients. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.24: RDD: Urban Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0010	-0.0066	-0.0811***	-0.114***	-0.0099	-0.0091	0.0561***	0.0548***
	(0.0132)	(0.0133)	(0.0296)	(0.0299)	(0.0163)	(0.0165)	(0.0199)	(0.0177)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	71	63	47	36	81	75	46	52
Bandwidth Right	71	83	47	69	81	82	46	72
Observations	2288	2356	1520	1696	2610	2531	1488	2000
Panel B: Fall ST								
Transition	-0.0035	-0.0076	0.0646***	0.0686***	-0.0189	-0.0221	-0.0263	-0.0257
	(0.0158)	(0.0169)	(0.0203)	(0.0243)	(0.0200)	(0.0222)	(0.0213)	(0.0218)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	46	34	84	58	54	38	45	37
Bandwidth Right	46	46	84	55	54	51	45	54
Observations	1442	1262	2688	1824	1744	1440	1456	1472

Note: The dependent variable is the log of the daily number of urban automobile accidents. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.25: DID Model: Urban Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0223**				
	(0.00930)				
First week of DST		-0.0116	-0.0516*	-0.0385*	0.0384*
		(0.0154)	(0.0288)	(0.0209)	(0.0201)
Second week of DST		-0.0176	-0.0739**	-0.0226	0.0266
		(0.0157)	(0.0293)	(0.0213)	(0.0205)
Remainder of Spring DST		-0.0264***	-0.0562***	-0.0444***	0.0109
		(0.0102)	(0.0191)	(0.0139)	(0.0133)
Fall ST	0.00807				
	(0.00902)				
First week of ST		-0.0033	0.0418	-0.0105	-0.0262
		(0.0147)	(0.0274)	(0.0199)	(0.0192)
Second week of ST		0.0252*	0.0722***	0.0059	0.0168
		(0.0149)	(0.0279)	(0.0202)	(0.0195)
Remainder of Fall ST		0.0079	0.0089	0.0115	0.0106
		(0.0102)	(0.0190)	(0.0138)	(0.0133)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.823	0.823	0.717	0.737	0.737

Note: The dependent variable is daily log of urban automobile crashes. The model estimated is a Poisson Pseudo maximum likelihood model. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.26: RDD: Rural Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	0.0211	0.0062	-0.0104	-0.0091	-0.0545*	-0.0529*	0.104**	0.0912**
	(0.0316)	(0.0285)	(0.0413)	(0.0434)	(0.0293)	(0.0310)	(0.0435)	(0.0393)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	50	49	81	74	87	71	39	45
Bandwidth Right	50	87	81	72	87	82	39	56
Observations	1616	2200	2610	2352	2808	2467	1264	1632
Panel B: Fall ST								
Transition	0.0195	-0.0051	-0.0011	-0.0173	0.159***	0.152***	-0.102***	-0.106***
	(0.0261)	(0.0249)	(0.0418)	(0.0409)	(0.0350)	(0.0345)	(0.0343)	(0.0352)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	60	93	56	69	64	71	60	54
Bandwidth Right	60	43	56	47	64	60	60	62
Observations	1876	2074	1808	1871	2064	2109	1936	1872

Note: The dependent variable is the log of the daily number of rural automobile accidents in columns 1 and 2, and the log of the daily number of rural automobile accidents plus one for the remaining columns. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.27: DID Model: Rural Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0060				
	(0.0181)				
First week of DST		0.0009	-0.0776	-0.0183	0.0383
		(0.0299)	(0.0557)	(0.0437)	(0.0387)
Second week of DST		0.0049	0.0462	-0.0888**	-0.0027
		(0.0305)	(0.0465)	(0.0442)	(0.0393)
Remainder of Spring DST		-0.0139	-0.0106	-0.102***	-0.0161
		(0.0198)	(0.0355)	(0.0280)	(0.0268)
Fall ST	0.0248				
	(0.0175)				
First week of ST		-0.0057	-0.0820*	0.0712**	-0.0470
		(0.0285)	(0.0427)	(0.0351)	(0.0428)
Second week of ST		0.0096	-0.0357	0.120***	-0.0692*
		(0.0290)	(0.0471)	(0.0386)	(0.0407)
Remainder of Fall ST		0.0422**	-0.0912***	0.196***	-0.0262
		(0.0198)	(0.0333)	(0.0278)	(0.0273)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.742	0.742	0.202	0.252	0.299

Note: The dependent variable the daily log of rural automobile crashes in columns 1 and 2, and the daily count for the remaining columns. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Columns 3, 4 and 5 use a Poisson count model (PPML) to estimate the coefficients. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.28: RDD: North Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	0.0109	-0.0009	-0.0207	-0.0217	-0.0316	-0.0340	0.0190	0.0108
	(0.0327)	(0.0275)	(0.0429)	(0.0431)	(0.0389)	(0.0374)	(0.0394)	(0.0369)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	42	45	62	59	62	54	50	49
Bandwidth Right	42	95	62	65	62	83	50	73
Observations	1360	2272	2000	2000	2000	2212	1616	1968
Panel B: Fall ST								
Transition	-0.0310	-0.0326	-0.0131	-0.0187	0.0028	0.0098	-0.0847**	-0.0860**
	(0.0302)	(0.0295)	(0.0434)	(0.0410)	(0.0415)	(0.0400)	(0.0372)	(0.0377)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	54	56	68	100	52	62	55	51
Bandwidth Right	54	58	68	61	52	50	55	57
Observations	1690	1784	2192	2564	1680	1808	1776	1744

Note: The dependent variable is the log of the daily number of automobile accidents in columns 1 and 2, and the log of the daily number of automobile accidents plus one for the remaining columns in the northern Chile. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.29: DID Model: North Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0223				
	(0.0194)				
First week of DST		-0.0186	-0.0417	-0.0157	-0.0062
		(0.0322)	(0.0505)	(0.0433)	(0.0409)
Second week of DST		-0.0382	-0.0280	-0.0700*	-0.0504
		(0.0328)	(0.0497)	(0.0425)	(0.0401)
Remainder of Spring DST		-0.0193	0.0246	-0.0422	-0.0206
		(0.0213)	(0.0351)	(0.0278)	(0.0259)
Fall ST	-0.0057				
	(0.0189)				
First week of ST		-0.0149	0.0049	0.0067	-0.0274
		(0.0307)	(0.0484)	(0.0431)	(0.0392)
Second week of ST		0.0094	0.0290	-0.0056	-0.0021
		(0.0312)	(0.0500)	(0.0375)	(0.0393)
Remainder of Fall ST		-0.0078	-0.0324	-0.0027	-0.0175
		(0.0213)	(0.0365)	(0.0290)	(0.0280)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.675	0.675	0.174	0.178	0.213

Note: The dependent variable the daily log of automobile in the northern Chile in columns 1 and 2 and the daily count for the remaining columns. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Columns 3, 4 and 5 use a Poisson count model (PPML) to estimate the coefficients. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.30: RDD: Center Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0093	-0.0192	-0.105***	-0.131***	-0.0190	-0.0294	0.0642***	0.0670***
	(0.0156)	(0.0152)	(0.0304)	(0.0309)	(0.0197)	(0.0196)	(0.0217)	(0.0205)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	61	54	50	39	70	63	46	53
Bandwidth Right	61	90	50	69	70	79	46	52
Observations	1968	2331	1616	1744	2224	2288	1488	1696
Panel B: Fall ST								
Transition	-0.0045	-0.0028	0.0672***	0.0671***	-0.0102	-0.00857	-0.0359	-0.0328
	(0.0175)	(0.0181)	(0.0224)	(0.0259)	(0.0227)	(0.0233)	(0.0241)	(0.0243)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	45	40	82	61	49	43	46	43
Bandwidth Right	45	46	82	56	49	51	46	50
Observations	1411	1352	2626	1888	1584	1520	1488	1504

Note: The dependent variable is the log of the daily number of automobile accidents in the central Chile. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.31: DID Model: Center Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0259**				
	(0.0103)				
First week of DST		-0.0098	-0.0794**	-0.0146	0.0415*
		(0.0171)	(0.0321)	(0.0236)	(0.0229)
Second week of DST		-0.0130	-0.0879***	-0.0056	0.0380
		(0.0174)	(0.0327)	(0.0241)	(0.0233)
Remainder of Spring DST		-0.0340***	-0.0841***	-0.0457***	0.0079
		(0.0113)	(0.0213)	(0.0156)	(0.0152)
Fall ST	0.0144				
	(0.00998)				
First week of ST		-0.0006	0.0406	-0.0021	-0.0279
		(0.0162)	(0.0306)	(0.0225)	(0.0218)
Second week of ST		0.0312*	0.0739**	0.0365	-0.0057
		(0.0165)	(0.0311)	(0.0229)	(0.0222)
Remainder of Fall ST		0.0166	0.0114	0.0344**	0.0063
		(0.0113)	(0.0212)	(0.0156)	(0.0151)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.696	0.697	0.603	0.574	0.602

Note: The dependent variable is daily log of automobile crashes in the central Chile. The model estimated is a Poisson Pseudo maximum likelihood model. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.32: RDD: South Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	0.0288	0.0255	0.0070	0.0030	0.0017	0.0028	0.0596*	0.0481
	(0.0256)	(0.0236)	(0.0358)	(0.0336)	(0.0323)	(0.0315)	(0.0357)	(0.0314)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	51	49	70	68	58	54	48	48
Bandwidth Right	51	80	70	96	58	73	48	91
Observations	1648	2081	2256	2657	1872	2048	1552	2252
Panel B: Fall ST								
Transition	0.0288	0.0329	0.0237	0.0077	0.0748**	0.0690**	-0.0179	-0.0197
	(0.0218)	(0.0238)	(0.0354)	(0.0330)	(0.0337)	(0.0330)	(0.0307)	(0.0298)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	77	63	73	94	65	78	58	56
Bandwidth Right	77	65	73	76	65	54	58	69
Observations	2394	2001	2347	2710	2096	2118	1872	2016

Note: The dependent variable is the log of the daily number of automobile accidents in the southern Chile in columns 1 and 2, and the log of the daily number of automobile accidents plus one for the remaining columns. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.33: DID Model: South Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0221				
	(0.0161)				
First week of DST		-0.0197	0.0076	-0.102***	0.0491
		(0.0267)	(0.0448)	(0.0290)	(0.0320)
Second week of DST		-0.0357	-0.0028	-0.0771**	0.0244
		(0.0272)	(0.0475)	(0.0301)	(0.0348)
Remainder of Spring DST		-0.0218	0.0022	-0.0718***	0.0101
		(0.0177)	(0.0273)	(0.0219)	(0.0223)
Fall ST	0.00941				
	(0.0156)				
First week of ST		-0.0105	-0.0212	0.0294	-0.0439
		(0.0254)	(0.0386)	(0.0336)	(0.0331)
Second week of ST		-0.0060	-0.0057	-0.0113	0.00582
		(0.0258)	(0.0366)	(0.0332)	(0.0350)
Remainder of Fall ST		0.0211	-0.0551*	0.0624***	0.0115
		(0.0176)	(0.0284)	(0.0228)	(0.0229)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.843	0.843	0.377	0.388	0.403

Note: The dependent variable the daily log of automobile crashes in the southern Chile in columns 1 and 2, and the daily count for the remaining columns. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Columns 3, 4 and 5 use a Poisson count model (PPML) to estimate the coefficients. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.34: RDD: Men Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	-0.0111	-0.0125	-0.0765***	-0.0836***	-0.0221	-0.0281	0.0546***	0.0548***
	(0.0129)	(0.0127)	(0.0268)	(0.0272)	(0.0181)	(0.0172)	(0.0208)	(0.0196)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	77	73	52	42	66	66	43	51
Bandwidth Right	77	93	52	64	66	88	43	48
Observations	2480	2686	1680	1712	2128	2489	1392	1600
Panel B: Fall ST								
Transition	0.004	0.0016	0.0444**	0.0494**	0.0120	0.0129	-0.0385*	-0.0350
	(0.0138)	(0.0163)	(0.0216)	(0.0243)	(0.0217)	(0.0218)	(0.0209)	(0.0213)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	63	41	75	63	48	47	48	40
Bandwidth Right	63	48	75	55	48	50	48	55
Observations	1969	1399	2409	1904	1552	1568	1552	1536

Note: The dependent variable is the log of the daily number of men automobile accidents. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.35: DID Model: Men Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0274***				
	(0.0090)				
First week of DST		-0.0158	-0.0627**	-0.0373*	0.0341*
		(0.0150)	(0.0278)	(0.0209)	(0.0205)
Second week of DST		-0.0235	-0.0681**	-0.0432**	0.0270
		(0.0153)	(0.0283)	(0.0213)	(0.0209)
Remainder of Spring DST		-0.0326***	-0.0519***	-0.0607***	0.0041
		(0.0099)	(0.0184)	(0.0138)	(0.0136)
Fall ST	0.0171*				
	(0.0088)				
First week of ST		-0.0027	0.0220	0.0026	-0.0256
		(0.0143)	(0.0264)	(0.0199)	(0.0195)
Second week of ST		0.0289**	0.0499*	0.0424**	0.0075
		(0.0145)	(0.0269)	(0.0202)	(0.0198)
Remainder of Fall ST		0.0218**	0.0067	0.0490***	0.0117
		(0.00992)	(0.0183)	(0.0138)	(0.0135)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.817	0.817	0.663	0.708	0.743

Note: The dependent variable is the log of daily men automobile crashes. The model estimated is a Poisson Pseudo maximum likelihood model. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.36: RDD: Women Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	0.0205	0.0259	-0.0513	-0.0579	0.0148	0.0121	0.0740**	0.0806***
	(0.0189)	(0.0227)	(0.0448)	(0.0420)	(0.0281)	(0.0325)	(0.0333)	(0.0309)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	88	60	49	44	76	51	48	48
Bandwidth Right	88	61	49	76	76	64	48	64
Observations	2841	1952	1584	1936	2448	1856	1552	1808
Panel B: Fall ST								
Transition	0.0114	0.0058	0.0928**	0.0959**	-0.0124	-0.0092	-0.0422	-0.0451
	(0.0246)	(0.0268)	(0.0370)	(0.0374)	(0.0347)	(0.0341)	(0.0341)	(0.0348)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	61	52	72	77	54	58	54	47
Bandwidth Right	61	51	72	61	54	54	54	59
Observations	1907	1612	2316	2215	1744	1808	1744	1712

Note: The dependent variable is the log of the daily number of women automobile accidents in columns 1 and 2, and the log of the daily number of automobile accidents plus one for the remaining columns. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.37: DID Model: Women Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0166				
	(0.0161)				
First week of DST		-0.0069	-0.0455	-0.0399	0.0777**
		(0.0267)	(0.0436)	(0.0334)	(0.0329)
Second week of DST		0.0058	-0.0057	0.0111	0.0162
		(0.0272)	(0.0428)	(0.0301)	(0.0361)
Remainder of Spring DST		-0.0242	-0.0710**	-0.0379*	0.0207
		(0.0177)	(0.0286)	(0.0218)	(0.0223)
Fall ST	-0.0124				
	(0.0156)				
First week of ST		-0.0105	0.0516	0.0133	-0.0279
		(0.0254)	(0.0370)	(0.0349)	(0.0331)
Second week of ST		0.00013	0.0969**	-0.0352	-0.0125
		(0.0258)	(0.0410)	(0.0350)	(0.0341)
Remainder of Fall ST		-0.0150	-0.0162	0.0295	-0.0102
		(0.0176)	(0.0276)	(0.0226)	(0.0223)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.741	0.741	0.350	0.293	0.270

Note: The dependent variable the daily log of women automobile crashes in columns 1 and 2, and the daily count for the remaining columns. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Columns 3, 4 and 5 use a Poisson count model (PPML) to estimate the coefficients. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.38: RDD: Drunk Driving Accidents

	All Hours		Morning		Evening		Least light impacted hours	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring DST								
Transition	0.0575	0.0380	-0.0072	-0.0185	-0.0416	-0.0342	0.0948*	0.0720
	(0.0452)	(0.0403)	(0.0428)	(0.0421)	(0.0495)	(0.0465)	(0.0533)	(0.0473)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	43	45	73	69	61	53	43	43
Bandwidth Right	43	83	73	82	61	90	43	82
Observations	1392	2068	2352	2435	1968	2315	1392	2019
Panel B: Fall ST								
Transition	0.0040	-0.0001	0.0455	0.0440	0.112**	0.108**	-0.0812*	-0.0796*
	(0.0390)	(0.0436)	(0.0464)	(0.0459)	(0.0502)	(0.0515)	(0.0484)	(0.0481)
Bandwidth Selector	mse	msetwo	mse	msetwo	mse	msetwo	mse	msetwo
Bandwidth Left	65	50	64	59	61	57	53	56
Bandwidth Right	65	55	64	79	61	61	53	51
Observations	2031	1646	2064	2224	1968	1904	1712	1728

Note: The dependent variable is the log of the daily number of drunk driving automobile accidents plus one. All specifications control for day-of-week, holiday, month and year FE, and they use a first-order polynomial. DST/ST is the estimate of the discontinuity in crashes that occurs immediately following the transition into DST/ST. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Bandwidth selectors are either a common mean squared error (MSE)-optimal bandwidth selector (denoted by one) or two different MSE-optimal bandwidth selectors below and above the cut-off (denoted by two). Both bandwidth selectors use a triangular kernel. Estimations include all days between 01/01/2002 and 12/31/2018. Observations change depending on the bandwidth. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D.39: DID Model: Drunk Driving Accidents

	All Hours		Morning	Evening	Least Light Impacted Hours
	(1)	(2)	(3)	(4)	(5)
Spring DST	-0.0442				
	(0.0294)				
First week of DST		0.0099	0.0453	-0.119*	0.0456
		(0.0396)	(0.0779)	(0.0704)	(0.0432)
Second week of DST		-0.0143	0.0532	-0.0780	-0.0159
		(0.0410)	(0.0779)	(0.0708)	(0.0517)
Remainder of Spring DST		-0.0768**	-0.0771	-0.147***	-0.0467
		(0.0337)	(0.0656)	(0.0491)	(0.0364)
Fall ST	0.0840***				
	(0.0292)				
First week of ST		0.0373	0.0816	0.217***	-0.0614
		(0.0443)	(0.0761)	(0.0678)	(0.0541)
Second week of ST		0.0285	0.106	0.110	-0.0373
		(0.0497)	(0.0852)	(0.0714)	(0.0580)
Remainder of Fall ST		0.126***	0.0780	0.224***	0.108***
		(0.0332)	(0.0648)	(0.0504)	(0.0369)
Observations	6,209	6,209	6,209	6,209	6,209
$Adj - R^2$	0.550	0.551	0.437	0.270	0.414

Note: The dependent variable is the number of daily drunk driving automobile crashes. All regressions use a Poisson count model (PPML) to estimate the coefficients. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Morning/Evening, I pick +/- three hours around the average hour of the sunrise/sunset between the day before and after the transition. Least light impacted hours are those more than three hours away from the average sunrise and sunset time at the spring transition in every location. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

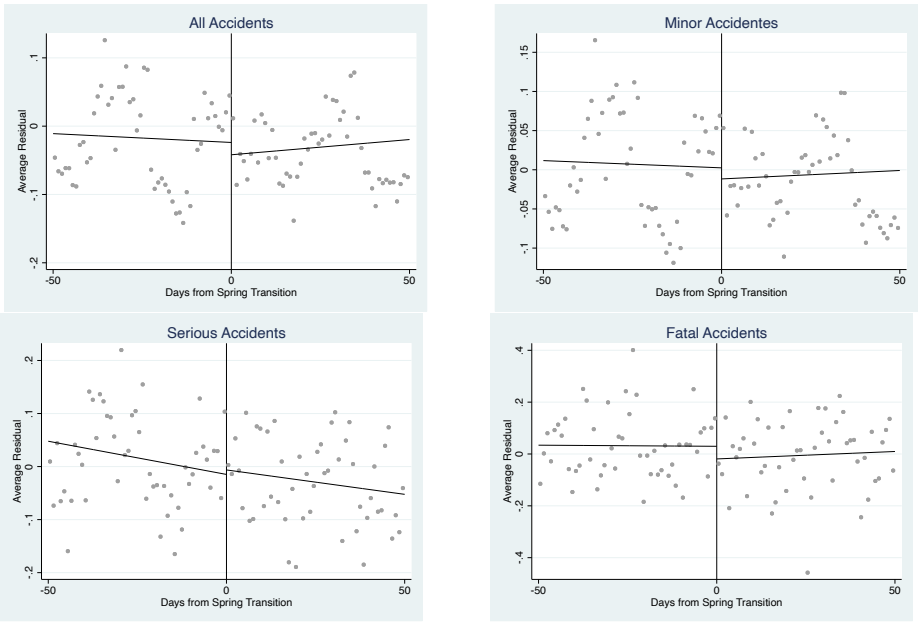
Table D.40: DID Model: Total Number of Victims by Severity

	Minor (1)	Serious (2)	Fatal (4)
First week of DST	-0.0287	-0.0140	0.110*
	(0.0219)	(0.0357)	(0.0582)
Second week of DST	0.00542	-0.0757**	-0.00339
	(0.0223)	(0.0364)	(0.0593)
Remainder of spring DST	-0.0332**	-0.0773***	-0.0744*
	(0.0145)	(0.0237)	(0.0386)
First week of ST	-0.0415**	-0.0580*	0.0466
	(0.0208)	(0.0340)	(0.0554)
Second week of ST	-0.00715	0.0357	0.0979*
	(0.0212)	(0.0346)	(0.0563)
Remainder of spring ST	-0.00160	0.000812	0.0727*
	(0.0145)	(0.0236)	(0.0385)
Observations	6,209	6,209	6,209
$Adj - R^2$	0.447	0.264	0.176

Note: The dependent variable is the log of the daily number of victims of automobile accidents by level of severity. All specifications use day-of-year, day-of-week, holiday, and year dummies. Remainder of spring DST (fall ST) is an indicator variable equal to one if the day occurs after the first two weeks of DST (ST) and after (before) July 1st. Estimations include all days between 01/01/2002 and 12/31/2018. Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1

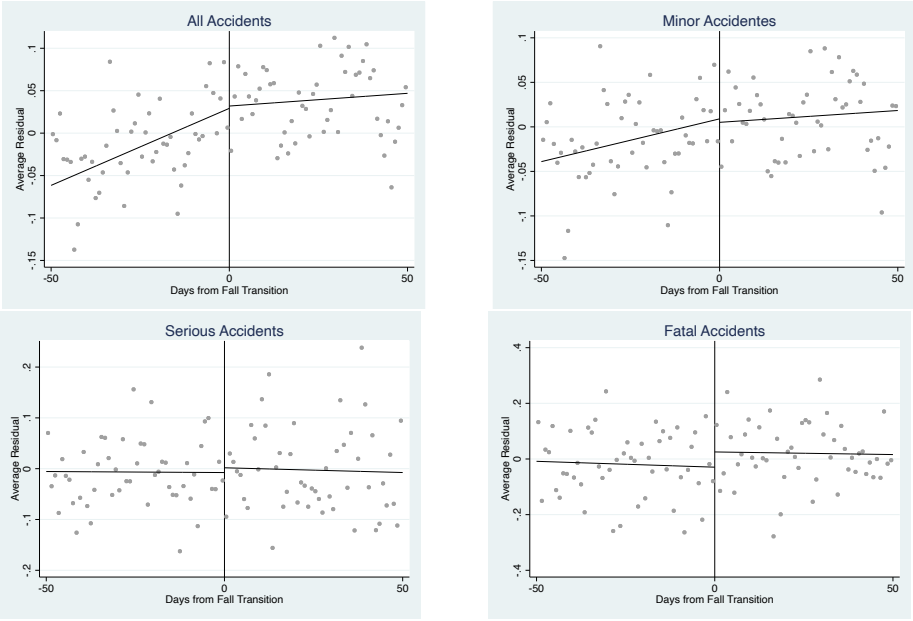
E Appendix: Figures

Figure E.1: Residual Plots from the Transition into DST by type of accidents



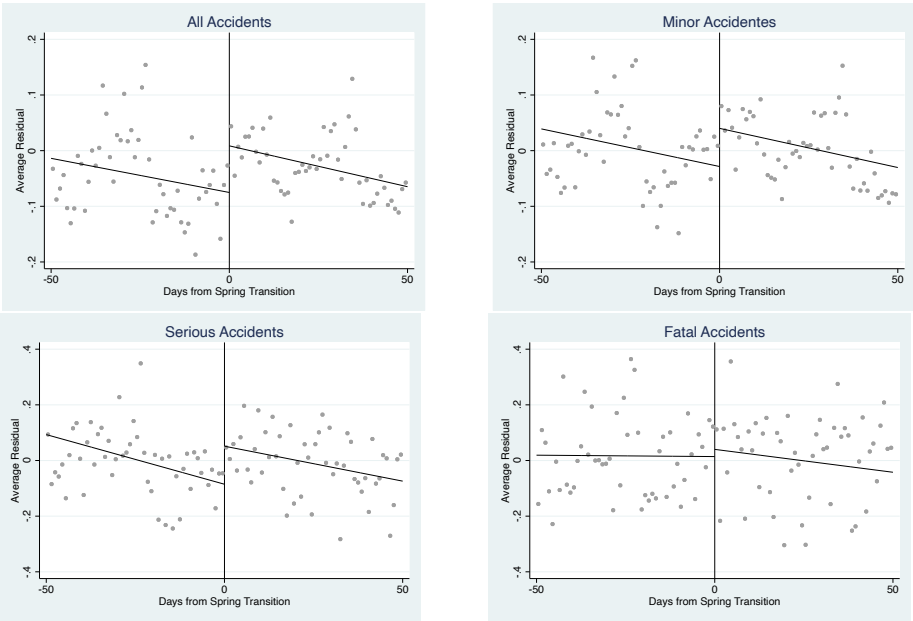
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.2: Residual Plots from the Transition into ST by type of accidents



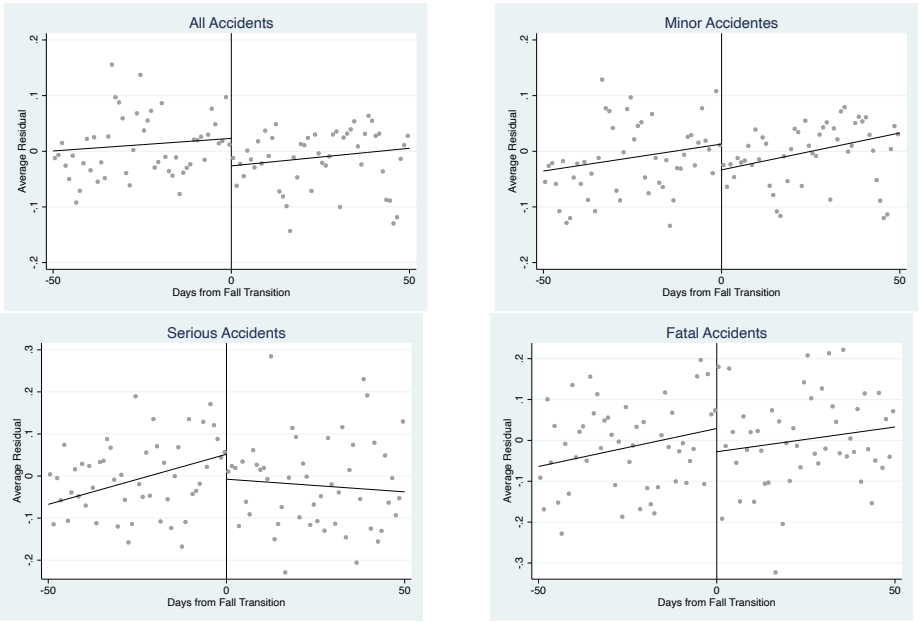
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.3: Residual Plots from the Transition into DST by type of accidents in the Least Light Impacted Hours



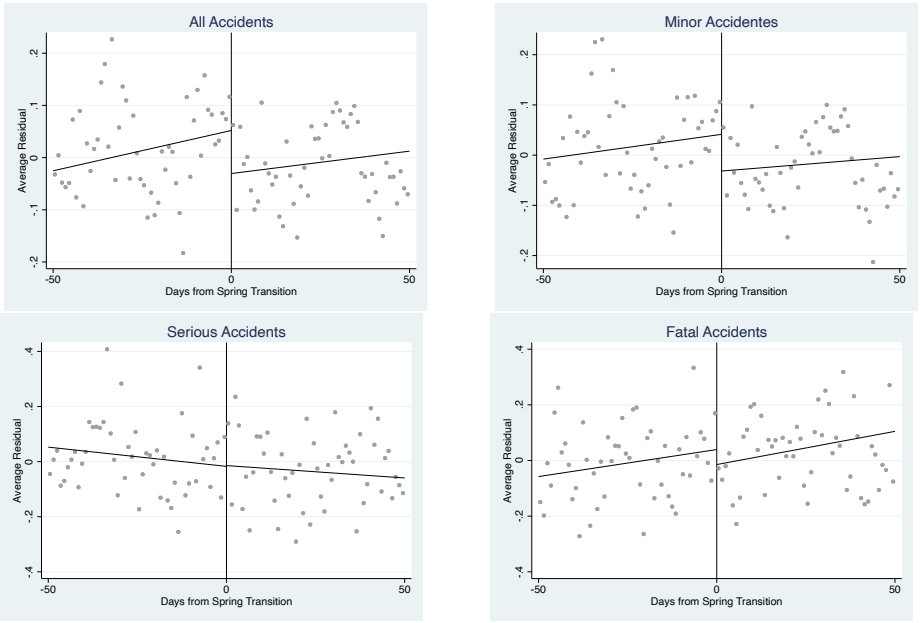
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.4: Residual Plots from the Transition into ST by type of accidents in the Least Light Impacted Hours



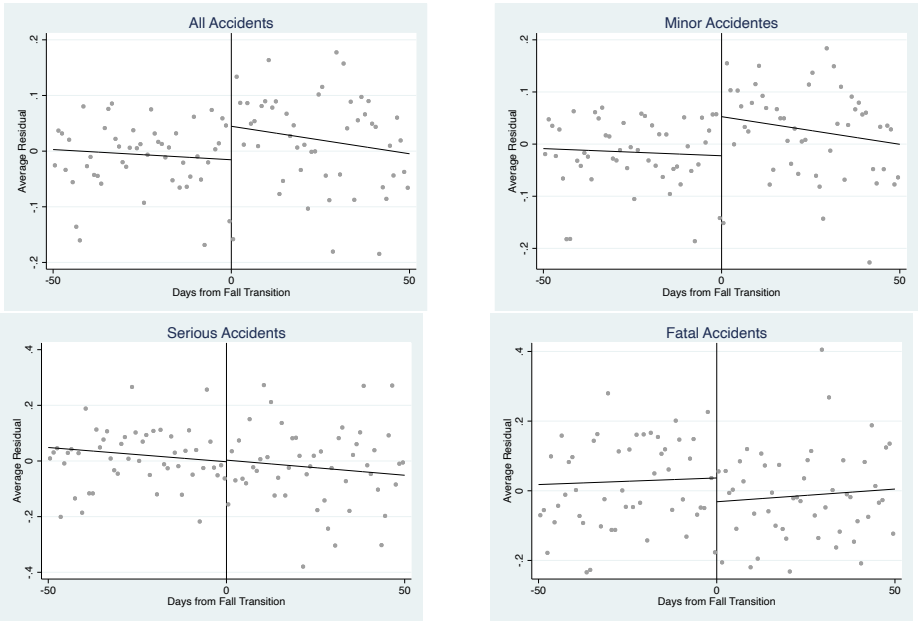
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.5: Residual Plots from the Transition into DST by type of accidents in the Morning



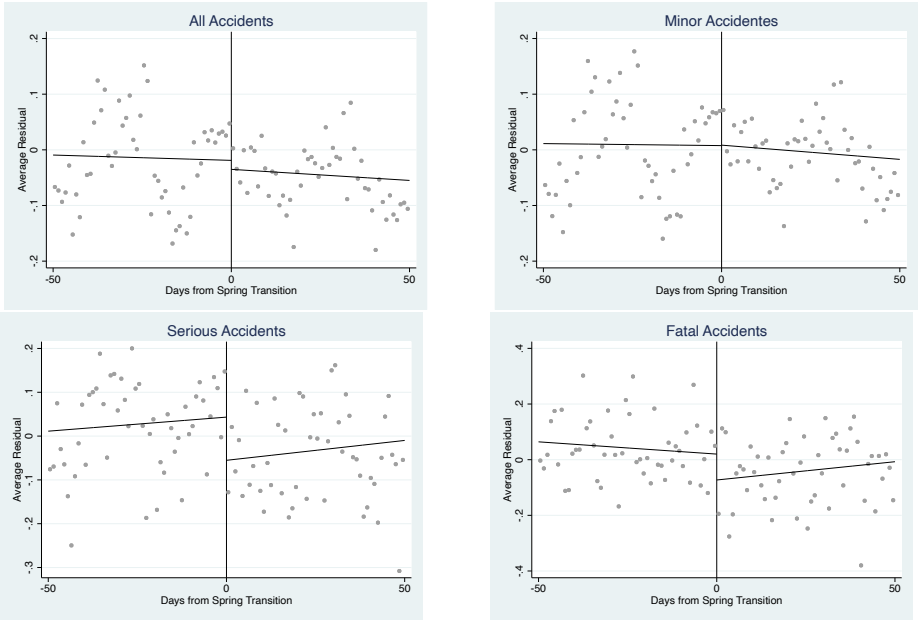
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.6: Residual Plots from the Transition into ST by type of accidents in the Morning



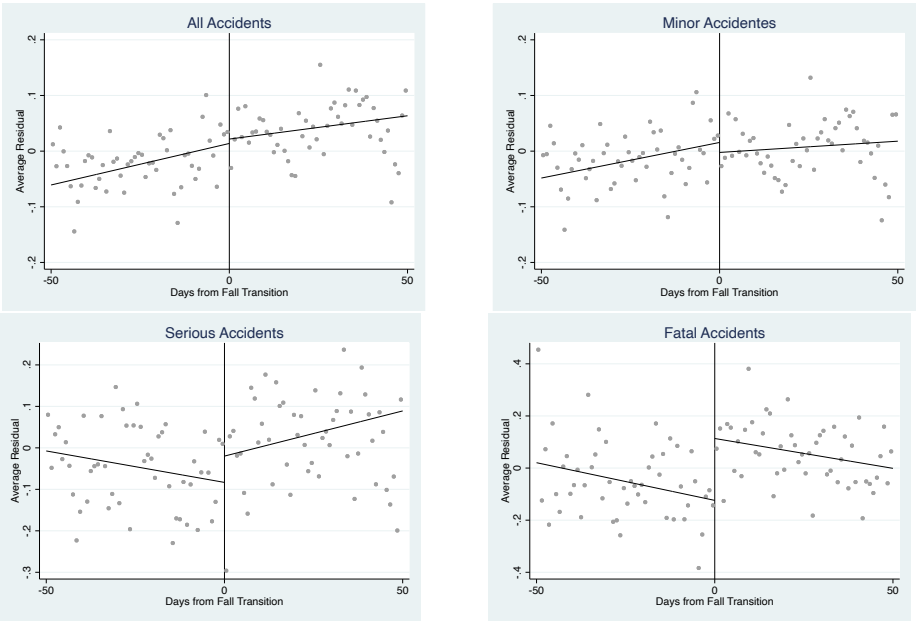
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.7: Residual Plots from the Transition into DST by type of accidents in the Evening



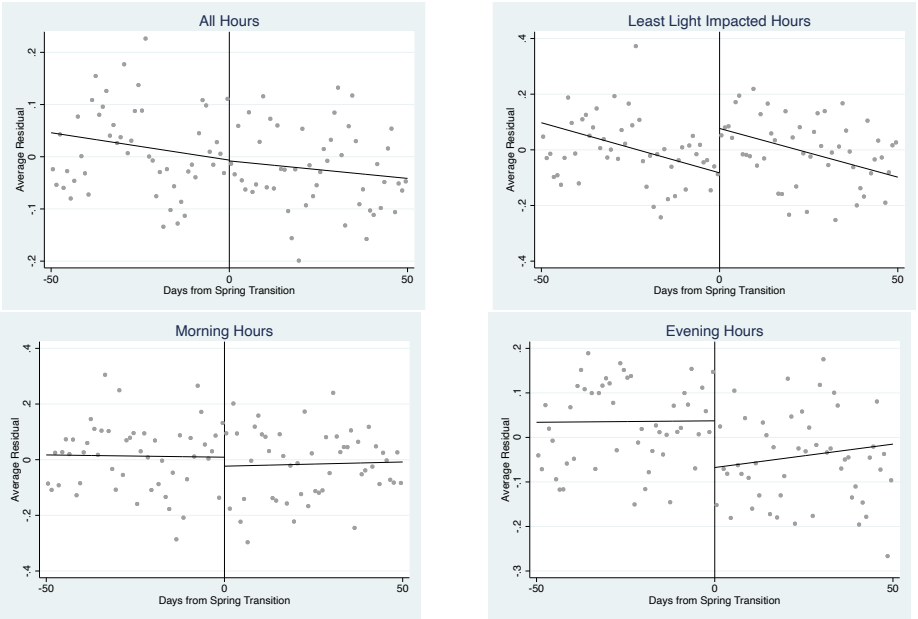
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.8: Residual Plots from the Transition into ST by type of accidents in the Evening



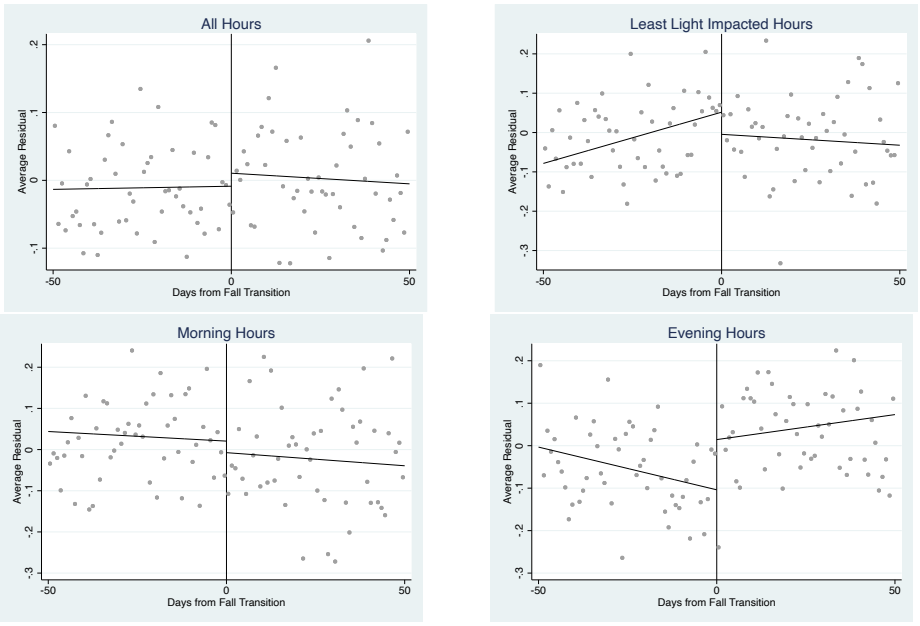
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.9: Residual Plots from the Transition into DST for Serious+Fatal Accidents



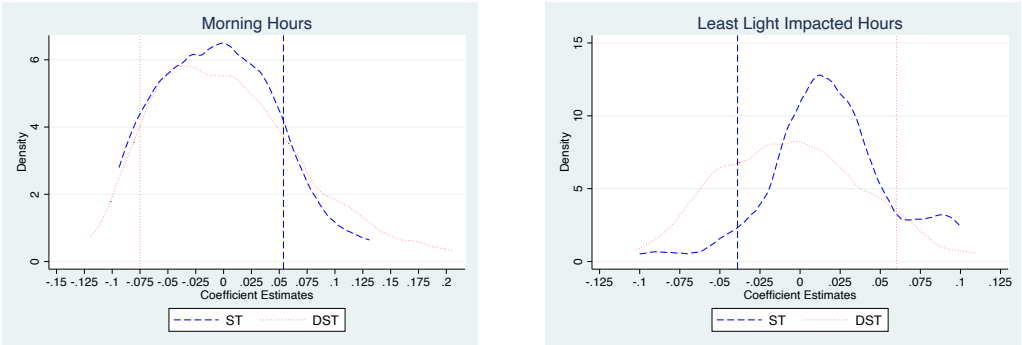
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.10: Residual Plots from the Transition into ST for Serious+Fatal Accidents



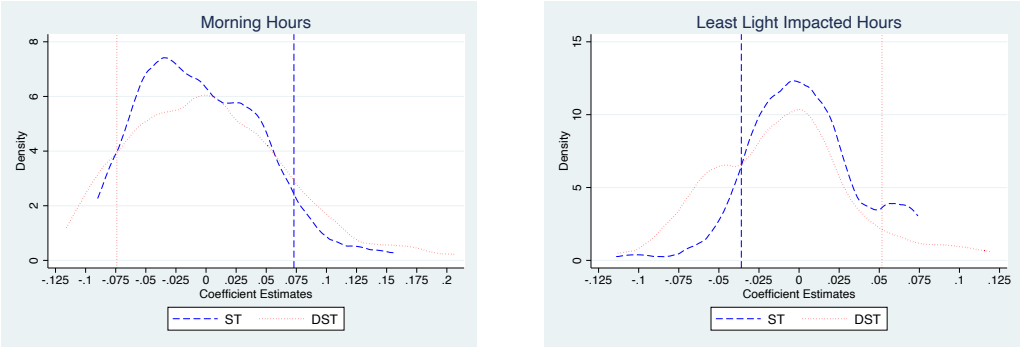
Note: The residuals are computed from a regression of natural log of accidents on day-of-week, month, year and holidays FE. Each point is the average of all residuals for that date relative to the transition. Fitted lines are results of locally weighted regressions using a triangular kernel.

Figure E.11: Permutation Test for the Least Light Impacted and Morning Coefficients: All Accidents



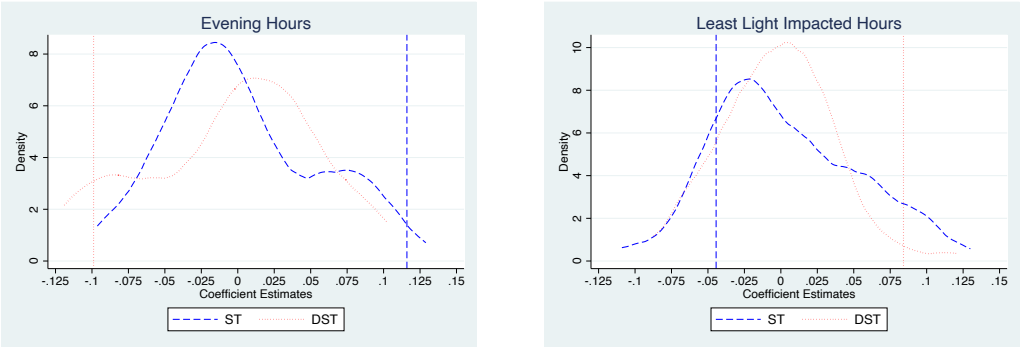
Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. ST means transition into ST and DST, transition into DST. Morning p-values are 0.12 (ST) and 0.09 (DST). Least Light Impacted Hours p-values are 0.06 (ST) and 0.05 (DST).

Figure E.12: Permutation Test for the Least Light Impacted and Morning Coefficients: Minor Accidents



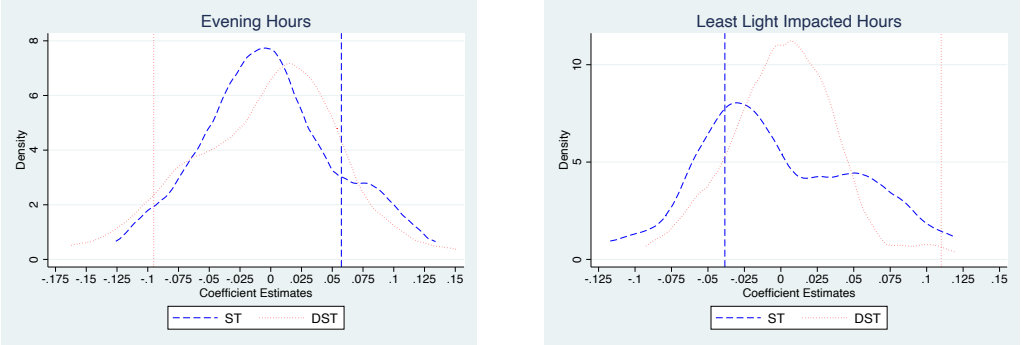
Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. ST means transition into ST and DST, transition into DST. Morning p-values are 0.06 (ST) and 0.1 (DST). Least Light Impacted Hours p-values are 0.07 (ST) and 0.07 (DST).

Figure E.13: Permutation Test for the Least Light Impacted and Evening Coefficients: Serious+Fatal Accidents



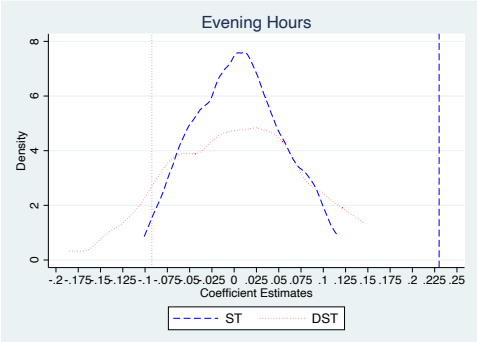
Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. ST means transition into ST and DST, transition into DST. Evening p-values are 0.00 (ST) and 0.06 (DST). Least Light Impacted Hours p-values are 0.15 (ST) and 0.01 (DST).

Figure E.14: Permutation Test for the Least Light Impacted and Evening Coefficients: Serious Accidents



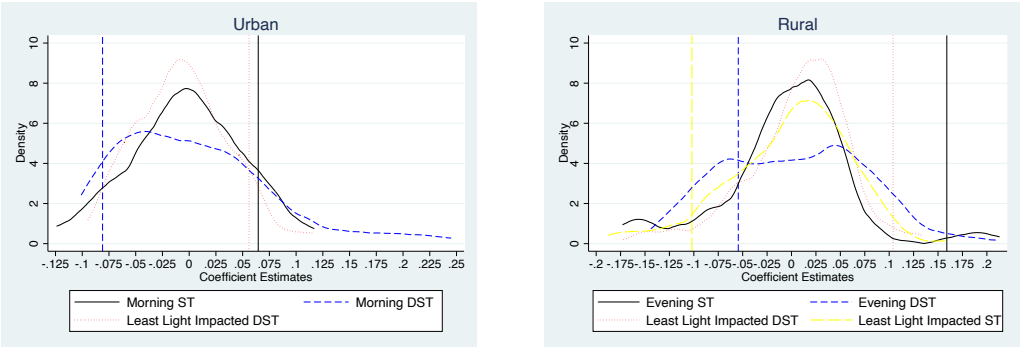
Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. ST means transition into ST and DST, transition into DST. Evening p-values are 0.18 (ST) and 0.07 (DST). Least Light Impacted Hours p-values are 0.14 (ST) and 0.00 (DST).

Figure E.15: Permutation Test for the Evening Coefficients: Fatal Accidents



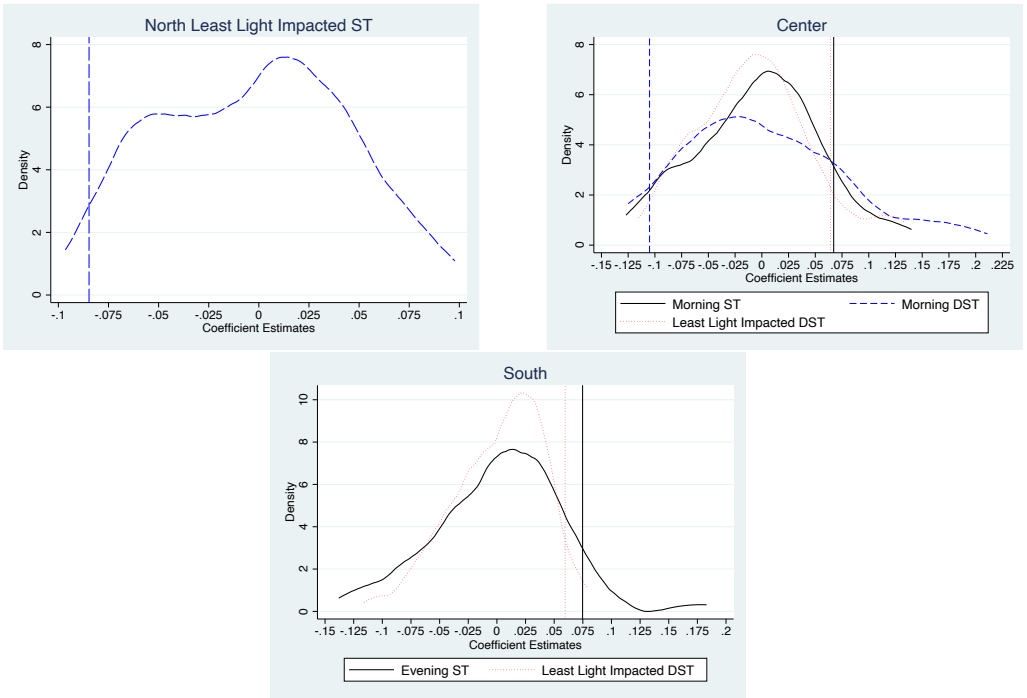
Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. ST means transition into ST and DST, transition into DST. Evening p-values are 0.00 (ST) and 0.13 (DST).

Figure E.16: Permutation Test for Estimated Coefficients by Zone



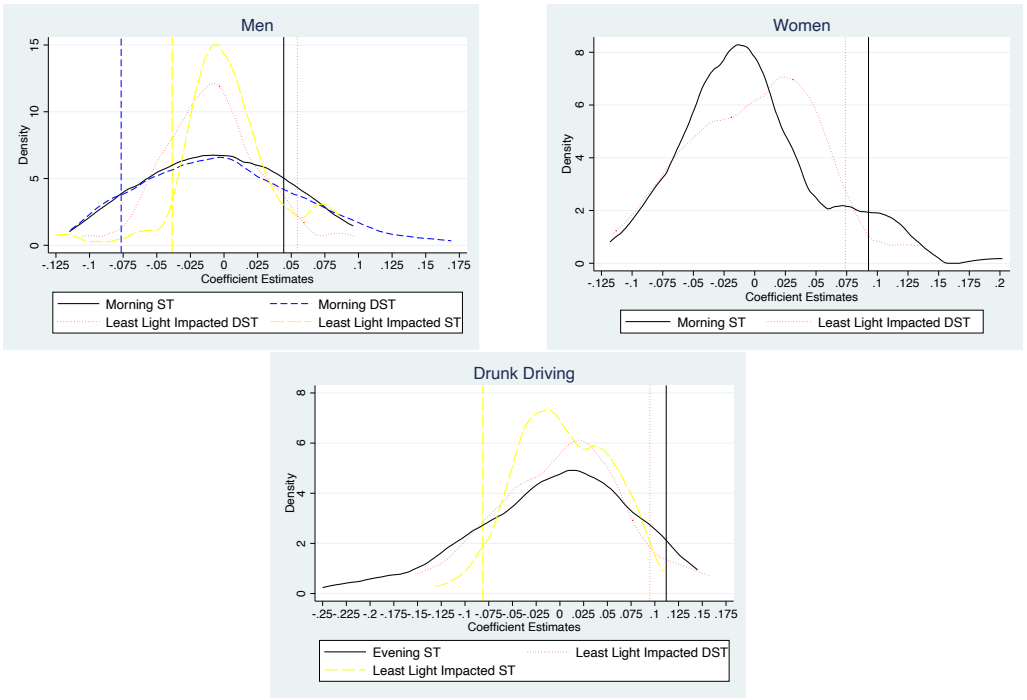
Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. Urban p-values are 0.09 (Morning ST), 0.08 (Morning DST) and 0.06 (Least Light Impacted DST). Rural p-values are 0.03 (Evening ST), 0.25 (Evening DST), 0.07 (Least Light Impacted ST) and 0.02 (Least Light Impacted DST).

Figure E.17: Permutation Test for Estimated Coefficients by Geographic Regions



Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. North p-value is 0.01 (Least Light Impacted ST). Center p-values are 0.08 (Morning ST), 0.05 (Morning DST) and 0.06 (Least Light Impacted DST). South p-values are 0.06 (Evening ST) and 0.03 (Least Light Impacted DST).

Figure E.18: Permutation Test for Estimated Coefficients by Gender and Drunk Driving



Note: The figures show the distribution of coefficients using fake transition dates. The fake transition dates are defined starting the day following the optimal bandwidth of the baseline regression one by one up to 120 days later. Vertical lines correspond to the estimated coefficients. Men p-values are 0.15 (Morning ST), 0.11 (Morning DST), 0.06 (Least Light Impacted ST) and 0.03 (Least Light Impacted DST). Women p-values are 0.07 (Morning ST) and 0.05 (Least Light Impacted DST). Drunk Driving p-values are 0.04 (Evening ST), 0.03 (Least Light Impacted ST) and 0.07 (Least Light Impacted DST)